Available online at www.sciencedirect.com
ScienceDirect
Comput. Methods Appl. Mech. Engrg. 372 (2020) 113411
Computer methods in applied mechanics and engineering
www.elsevier.com/locate/cma

Novel sparseness-inducing dual Kalman filter and its application to tracking time-varying spatially-sparse structural stiffness changes and inputs

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Received 18 March 2020; received in revised form 25 July 2020; accepted 31 August 2020
Available online 28 September 2020

Abstract

While the theory of Bayesian system identification provides a probabilistic means for reliably and robustly inferring models of a dynamic system and their parameters based on measured dynamic response, exploiting sparseness during online tracking of changing model parameters is not well understood. The focus in this study is to implement the dual Kalman filter for real-time Bayesian sequential state and parameter identification based on noisy sensor signals while incorporating sparse Bayesian learning to impose sparse model parameter changes from their initial reference values. We also want out model to be able to capture the evolution of sparse model parameter changes between two successive time instants. To this end, we present a hierarchical Bayesian model for tracking the joint posterior distribution of the state and model parameter vectors for a monitored dynamical system, where the two afore-mentioned sparseness constraints (sparse changes from reference values and with time) are also effectively incorporated for each time. We show our stochastic model of the structural dynamical system can be represented as a coupled conditionally-linear Gaussian state space model for the state and model parameter vectors, leading to some interesting analytical properties of the method that allow quantities of interest to be calculated in real time by using Kalman filtering equations. The parameters for the measurement and state prediction errors are learned solely from the available data up to the current time and so our method resolves the well-known instability problem in Kalman filtering due to arbitrary assignment of the error-distribution parameters. Finally, two illustrative applications are presented, one for identification of stiffness degradation and the other for input time–history identification where both are based on noisy dynamic response measurements from a structure.
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Keywords: Bayesian system identification; Sparse Bayesian learning; Kalman filter; Online model error and noise identification; Dual sequential state and parameter estimation; Input time–history identification

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https://doi.org/10.1016/j.cma.2020.113411
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1. Introduction

In the last few decades, much effort has been devoted to the subject of real-time state estimation of models for dynamic system based on a series of measurements over time, and exploiting prior information about the temporal regularity of the output signals. There are well developed algorithms to manage both linear (e.g. the Kalman filter [1]) and nonlinear state–space models (e.g. extended Kalman filter (EKF) [2]); the particle/sequential-Monte-Carlo filter [3,4]; and the unscented Kalman filter [5–7]. These real-time state estimations methods are powerful tools and the applications are numerous in science and technology: control engineering (e.g., [8]), robot vision (e.g., [9]), signal processing (e.g., [10]), biology (e.g., [11]), power systems (e.g., [12]), economics (e.g., [13]) and thermal science (e.g., [14]), etc.

A real-time state estimation algorithm may be used to provide estimates of the whole state of a dynamic system (e.g., knowledge of the system displacements and velocities at all degrees of freedom of a model of a structural system), because it is often difficult or infeasible to measure the entire state. This estimation is conditional on a specified model and available input and output measurements. In practice, there is also a need to estimate time-varying model parameters based on noisy input and output measurements, because appropriate values are not known initially and they also change slowly over the lifetime of the system. As a result, there has been a long interest in combined state and parameter estimation for dynamic systems. In Bayesian system identification [15], Bayesian filters have been proposed for simultaneous tracking of the system states and model parameters (e.g., [4,16]). Since an accurate model of the true structural system is usually not known beforehand, these methods allow the models to be adapted to better match the true system's behavior. Probabilistically speaking, the objective is to find the time-varying joint posterior distribution of the system states and model parameters conditional on the measured system signals.

There are two common approaches to meet the goal: joint Kalman filter [17] and dual Kalman filter [18]. In the *joint Kalman filter*, the state and parameter vectors are combined, and a Kalman filter estimates the posterior PDF (probability density function) of this augmented state vector. This allows direct modeling of the correlations between system states and model parameters. However, even for linear dynamic systems, this joint approach results in nonlinear dynamics over the augmented state vector so only approximate numerical methods are available. In addition, this method has the disadvantage of involving large matrix operations due to the high dimensionality of the resulting augmented state vector. In the *dual Kalman filter*, two interacting but distinct Kalman filters are used for each of the state and parameter vector estimations. One filter computes the state posterior given a parameter estimate, while the other computes the parameter posterior given an estimate of the hidden states. The two separate Kalman filters interact by each feeding its estimate into the other. The computational complexity is smaller and the matrix operations may be numerically better conditioned. However, by decoupling model parameters from states, the inference of their joint posterior PDF is a nontrivial task. Furthermore, there are some inherent difficulties for reliable online state-parameter estimation in real applications.

One of the main difficulties lies in the fact that the quality of the Kalman filter estimation depends heavily on the selection of uncertain parameters specifying the probability models for the dynamic model errors and measurement noise in the state–space model (e.g., the parameters for the prediction error covariance). In most of earlier work, they are treated as 'tuning parameters' and adjusted in an ad-hoc manner while carrying out the estimation. An improper choice of prediction error characteristics leads to deterioration in time of the performance of the Kalman filer algorithms and, in the worst case, the estimators may diverge [19,20]. Some researchers have estimated these covariance parameters from data. One approach to do this is the auto-covariance least-squares technique that uses the time-lagged auto-covariances between the measurements at different times [21]. Another approach is that of Bavdekar et al. [22], who developed maximum likelihood estimates (MLE) of the noise covariance matrices from operating data. On the other hand, Yuen et al. [23,24] proposed a Bayesian method for estimation of the noise parameters in the Extended Kalman filter. The computations employed the finite difference method to approximate required derivatives. In this study, analytical solutions of the MAP (maximum a posteriori) estimates of the time-varying noise parameters are derived as a model class assessment problem [15,25], and it is suitable for online implementation.

Another fundamental difficulty is that Kalman filters for dynamic systems typically suffer from a lack of observability when the number of measurements is limited; however, it is not easy to integrate additional prior knowledge, e.g., model structure sparseness, into the Kalman filter formulation to counteract insufficient measurements. For static models, exploiting sparseness in linear expansions has received significant attention in

recent studies of inverse problems, especially in the area of compressed sensing (e.g., [26–31]), where inducing low-dimensional structure through a judicious choice of regularizers or priors has been enormously successful in overcoming insufficient measurements.

There are many applications where one can reasonably expect significant potential gains from jointly leveraging both the sparseness and temporal regularization structures in a Kalman filter framework. The sequential estimation of dynamically changing sparse states by leveraging recent advances in sparse approximation [32,33] has recently captured the attention of various researchers, although the focus is mostly on tracking the sparse signals in compressive sensing [34–37]. Along with a sparseness constraint on the state vector itself, these methods also aim to capture the evolution of sparse changes with shared "common sparseness", i.e., the changes between two successive time instants are also sparse. This is an important constraint that is often appropriate to make to aid estimation at the current time, and that is expected to reduce the uncertainty when estimating states that are undergoing a change. In the Bayesian community, Sejdinovic et al. [38] proposed a sequential Monte Carlo sampling approach for sequential compressed sensing of dynamically changing sparse signals where they recursively estimate the full posterior distribution at each time instant. Huang et al. [39] proposed a dynamic sparseness embedded Kalman filter algorithm for sequential Bayesian compressive sensing. However, it is only applicable for time-varying state estimation and simultaneous tracking of the system states and model parameters cannot be achieved.

This work involves combining sparse Bayesian learning [40] with dual Kalman filters by employing a hierarchical probabilistic model. We utilize two inference levels within this hierarchy, namely, state and parameter estimation and noise-parameter learning. A hierarchical modeling idea was also used in a sequential data environment in [41] but no dynamic sparseness constraint was imposed in their method. One application presented in this paper treats the problem of estimating the unknown states and parameters of linear dynamic structural systems that undergo degrading stiffness during their operational life. The state estimation itself is a secondary goal compared to that of estimating the slowly time-varying structural model parameters, as the objective is to estimate the structural damage accumulation. We have proposed previously to use a sparse Bayesian perspective to tackle the stiffness loss inverse problem based on dynamic modal data [30,31,42–44], but the algorithms presented were not formulated for real-time online application. The other application in this paper is structural input identification, e.g., structural impact inputs, together with state estimation. Very few authors have investigated the online tracking of the dynamic sparseness of stiffness losses from damage and deterioration along with inferring structural inputs.

The remainder of the paper is organized as follows. Section 2 is devoted to a formulation of the state–space equations of a bi-linear dynamical system. Section 3 introduces how dynamic sparseness can be embedded in a dual Kalman filter formulation for estimation of both the unknown state and model parameters. The corresponding pseudo-code of the algorithm is also presented. Numerical investigations of the performance of the proposed algorithm using two illustrative examples are presented in Section 4, while conclusions and future work are discussed in Section 5.

2. Stochastic state–space model of dynamic system

Let $\mathbf{x}_i \in \mathbb{R}^{M_i}$ be the system model state vector and $\boldsymbol{\theta}_i \in \mathbb{R}^{N_i}$ be the model parameter vector for a model characterizing a linear dynamical system. Consider the following stochastic state–space equations for $\mathbf{x}_i$ and $\boldsymbol{\theta}_i$ for a sequence of discrete time instants t_i:

$$\begin{aligned} &\forall i \in \mathbb{Z}^+, \mathbf{x}_i = f_{i-1}\left(\mathbf{x}_{i-1}, \boldsymbol{\theta}_{i-1}\right) = \mathbf{P}_{i-1}\left(\boldsymbol{\theta}_{i-1}\right)\mathbf{x}_{i-1} + \mathbf{p}_{i-1}\left(\boldsymbol{\theta}_{i-1}\right) + \boldsymbol{\varepsilon}_i \\ &\boldsymbol{\theta}_i = \boldsymbol{\theta}_{i-1} + \boldsymbol{\delta}_i \end{aligned} \tag{1}$$

where $\mathbf{P}_{i-1} \in \mathbb{R}^{M_i \times M_{i-1}}$ and $\mathbf{p}_{i-1} \in \mathbb{R}^{M_i \times 1}$ are specified functions of uncertain model parameters $\boldsymbol{\theta}_{i-1}$; and $\boldsymbol{\delta}_i \in \mathbb{R}^{N_i}$ and $\boldsymbol{\varepsilon}_i \in \mathbb{R}^{M_i}$ are prediction error terms, commonly called process noise, in the model for the evolution of the parameter vector $\boldsymbol{\theta}_i$ and the system state vector $\mathbf{x}_i$, respectively. Their prior distributions are chosen as zero-mean Gaussian $\boldsymbol{\varepsilon}_i \sim \mathcal{N}\left(\mathbf{0}, \beta^{-1} q_i \mathbf{I}_{M_i}\right)$ and $\boldsymbol{\delta}_i \sim \mathcal{N}\left(0, \beta^{-1}\boldsymbol{\Lambda}_i\right)$, $\boldsymbol{\Lambda}_i = \operatorname{diag}\left(\lambda_{i,1}, \ldots, \lambda_{i,N_i}\right)$, based on the Principle of Maximum Information Entropy subject to the first two moment constraints [15].

We wish to induce sparseness in the model parameters in two ways. First, we want the changes $\boldsymbol{\theta}_i - \boldsymbol{\theta}_{i-1} = \boldsymbol{\delta}_i$ to be a sparse vector. The prior $\boldsymbol{\delta}_i \sim \mathcal{N}\left(0, \beta^{-1}\boldsymbol{\Lambda}_i\right)$ is inspired by Automatic Relevance Determination (ARD) prior in [40] and [45] because it can produce model sparseness effectively [40]. Each individual hyper-parameter $\lambda_{i,n}$ is associated independently with each $\delta_{i,n}$, thereby moderating the strength of the prior soft constraint on $\boldsymbol{\delta}_{i,n}$. Note

that as $\lambda_{i,n} \to 0$, the prior for $\delta_{i,n}$, and hence its posterior, approaches a Dirac delta-function at zero. Based on past work, if sparseness in $\boldsymbol{\delta}_i$ is appropriate for the model representation, then many $\lambda_{i,n}{}'s$ will approach zero during the learning process, forcing the corresponding $\delta_{i,n}{}'s$ to approach zero.

The second way we wish to induce sparseness is to discourage any changes of model parameter vector $\boldsymbol{\theta}_i$ compared with a specified reference vector $\boldsymbol{\theta}_i^c$; that is, $\Delta\boldsymbol{\theta}_i = \boldsymbol{\theta}_i - \boldsymbol{\theta}_i^c$ is also to be sparsely distributed. Inspired again by the ARD prior, we choose the prior $\Delta\boldsymbol{\theta}_i \sim \mathcal{N}\left(0, \beta^{-1}\mathbf{A}_i\right)$, $\mathbf{A}_i = \text{diag}\left(\alpha_{i,1}, \ldots, \alpha_{i,N_i}\right)$. Note that the hyper prior PDFs for q_i, $\boldsymbol{\lambda}_i = \left[\lambda_{i,1}, \ldots, \lambda_{i,N_i}\right]^T$ and $\boldsymbol{\alpha}_i = \left[\alpha_{i,1}, \ldots, \alpha_{i,N_i}\right]^T$ are simply taken as uniform over large open intervals that start at zero.

Our goal is to develop a Kalman filter formulation that simultaneously finds sparse solutions for both vectors $\boldsymbol{\delta}_i$ and $\Delta\boldsymbol{\theta}_i$ at each time instant. Assume incomplete noisy measurements $\mathbf{y}_i \in \mathbb{R}^{K_i}$ are observed for the system state vector $\mathbf{x}_i$ and model parameter vector $\boldsymbol{\theta}_i$ for each time instant t_i, and that the observation model can be expressed bi-linearly with respect to $\mathbf{x}_i$ and $\boldsymbol{\theta}_i$, respectively:

$$\mathbf{y}_i = h_i\left(\mathbf{x}_i, \boldsymbol{\theta}_i\right) + \mathbf{e}_i = \mathbf{C}_i\left(\boldsymbol{\theta}_i\right)\mathbf{x}_i + \mathbf{c}_i\left(\boldsymbol{\theta}_i\right) + \mathbf{e}_i = \mathbf{B}_i\left(\mathbf{x}_i\right)\boldsymbol{\theta}_i + \mathbf{b}_i\left(\mathbf{x}_i\right) + \mathbf{e}_i \tag{2}$$

where $\mathbf{C}_i \in \mathbb{R}^{K_i \times M_i}$ and $\mathbf{B}_i \in \mathbb{R}^{K_i \times N_i}$ are measurement matrices as a function of $\boldsymbol{\theta}_i$ and $\mathbf{x}_i$, respectively; $\mathbf{e}_i$ represents the unknown prediction error due to the observation model for specified system state vector $\mathbf{x}_i$ and model parameter vector $\boldsymbol{\theta}_i$ plus measurement errors; it is modeled with a zero-mean Gaussian distribution with common precision parameter β. This probability model gives the largest uncertainty for $\mathbf{e}_i$ (i.e. maximizes Shannon's information entropy [46](Jaynes, 1957)) subject to the first two moment constraints: $\mathbf{E}\left[\mathrm{e}_{i,k}\right] = 0, \mathbf{E}\left[\mathrm{e}_{i,k}^2\right] = \beta^{-1}, k = 1, \ldots, K_i$. A Gamma prior is assigned for the precision parameter β, i.e., $p\left(\beta|a_0, b_0\right) = \text{Gamma}\left(\beta|a_0, b_0\right)$, where the shape and rate parameters are taken as: $a_0 = 1$ and $b_0 = 10^{-10}$. The prior is almost flat since the mean $\mathbf{E}\left(\beta\right) = 10^{10}$ and variance $\text{Var}\left(\beta\right) = 10^{20}$. Note that the two coupled linear-in-the-parameter equations in (2) can be utilized to perform Kalman filtering for $\boldsymbol{\theta}_i$ and $\mathbf{x}_i$ separately; although the observation model for $\mathbf{y}_i$ is a nonlinear function of the augmented state vector $\boldsymbol{\zeta}_i = \left[\mathbf{x}_i^T, \boldsymbol{\theta}_i^T\right]^T$.

Remark 1. For linear structural dynamics problem, the structural displacement and velocity vectors for the time instant t_i can constitute the system state vector $\mathbf{x}_i$, and its evolution over time is typically formulated using the well-known continuous-time second-order differential equation. Usually in system identification, the structural stiffness parameters for each time instant t_i are of interest and so they would be included in the model parameter vector $\boldsymbol{\theta}_i$. The stiffness reduction from any damage usually occurs at only a few locations and so the stiffness change from the undamaged state is sparse. In addition, the occurrence of damage is infrequent and so stiffness changes between adjacent time stages should also be sparse. Based on a series of measured structural responses at each time instant, near real-time stiffness identification can be achieved by using the Kalman filter method presented in the paper. Our method is also applicable for identifying structural inputs, such as impact inputs, as we show in a later example.

3. Sparseness-inducing dual Kalman filter

3.1. Hierarchical sparse Bayesian modeling

The specified stochastic state–space model in Section 2 defines a "hidden" Markov chain for the state time history $\{\mathbf{x}_i\}_{i=0}^I$ and model parameter time history $\{\boldsymbol{\theta}_i\}_{i=0}^I$ by implying state transition PDFs:

$$p\left(\mathbf{x}_i|\mathbf{x}_{i-1}, \boldsymbol{\theta}_{i-1}, q_i, \beta\right) = \mathcal{N}\left(\mathbf{x}_i|\mathbf{P}_{i-1}\mathbf{x}_{i-1} + \mathbf{p}_{i-1}, \beta^{-1}q_i\mathbf{I}_{M_i}\right) \tag{3}$$

$$p\left(\boldsymbol{\theta}_i|\boldsymbol{\theta}_{i-1}, \boldsymbol{\lambda}_i, \beta\right) = \mathcal{N}\left(\boldsymbol{\theta}_i|\boldsymbol{\theta}_{i-1}, \beta^{-1}\boldsymbol{\Lambda}_i\right) \tag{4}$$

along with the state-to-output PDF:

$$p\left(\mathbf{y}_i|\mathbf{x}_i, \beta, \mathbf{y}_{0:\,i-1}\right) = \mathcal{N}\left(\mathbf{y}_i|\mathbf{C}_i\left(\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^-\right)\mathbf{x}_i + \mathbf{c}_i\left(\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^-\right), \beta^{-1}\mathbf{I}_{K_i} + \beta^{-1}\mathbf{B}_i\left(\mathbf{x}_i\right)\mathbf{E}_{\boldsymbol{\theta}_i}^-\left(\mathbf{B}_i\left(\mathbf{x}_i\right)\right)^T\right) \tag{5a}$$

$$p\left(\mathbf{y}_i|\boldsymbol{\theta}_i, \beta, \mathbf{y}_{0:\,i-1}\right) = \mathcal{N}\left(\mathbf{y}_i|\mathbf{B}_i\left(\boldsymbol{\mu}_{\mathbf{x}_i}^-\right)\boldsymbol{\theta}_i + \mathbf{b}_i(\boldsymbol{\mu}_{\mathbf{x}_i}^-), \beta^{-1}\mathbf{I}_{K_i} + \beta^{-1}\mathbf{C}_i\left(\boldsymbol{\theta}_i\right)\mathbf{E}_{\mathbf{x}_i}^-\left(\mathbf{C}_i\left(\boldsymbol{\theta}_i\right)\right)^T\right) \tag{5b}$$

which are derived in Appendix A. In addition, we choose the reference value $\boldsymbol{\theta}_i^c$ as pseudo-data [42] to define the likelihood function for $\boldsymbol{\theta}_i$ given $\boldsymbol{\theta}_i^c$ as:

$$p\left(\boldsymbol{\theta}_i^c|\boldsymbol{\theta}_i, \boldsymbol{\alpha}_i\right) = \prod_{n=1}^{N_i}\mathcal{N}\left(\theta_{i,n}^c|\theta_{i,n}, \alpha_{i,n}\right) = \mathcal{N}\left(\boldsymbol{\theta}_i^c|\boldsymbol{\theta}_i, \beta^{-1}\mathbf{A}_i\right) \tag{6}$$

where $\mathbf{A}_i = \text{diag}\left(\alpha_{i,1}, \ldots, \alpha_{i,N_i}\right)$ as before. This provides a soft constraint that it is consistent with the sparseness prior on $\Delta\boldsymbol{\theta}_i = \boldsymbol{\theta}_i - \boldsymbol{\theta}_i^c$. Note that in order to provide a tractable updating of the prediction error precision parameter β, we incorporate β in the state transition PDFs in (3), (4), (5) and (6).

3.2. Bayesian inference

For the initial time instant ($i = 0$), we choose $p\left(\boldsymbol{\theta}_0\right)$ and $p\left(\mathbf{x}_0\right)$ as Gaussians $\mathcal{N}\left(\boldsymbol{\theta}_0|\boldsymbol{\mu}_{\boldsymbol{\theta}_0}^+, \beta^{-1}\mathbf{E}_{\boldsymbol{\theta}_0}^+\right)$ and $\mathcal{N}\left(\mathbf{x}_0|\boldsymbol{\mu}_{\mathbf{x}_0}^+, \beta^{-1}\mathbf{E}_{\mathbf{x}_0}^+\right)$with covariance matrices:

$$\mathbf{E}_{\boldsymbol{\theta}_0}^+ = \mathbf{I}_{N_0}, \mathbf{E}_{\mathbf{x}_0}^+ = \mathbf{I}_{M_0} \tag{7}$$

We approximate the posterior PDFs for $\boldsymbol{\theta}_{i-1}$ and $\mathbf{x}_{i-1}$ conditional on the data up to the $(i-1)$ th time instant by Gaussian PDFs:

$$p\left(\boldsymbol{\theta}_{i-1}|\boldsymbol{\theta}_{0:\,i-1}^c, \mathbf{y}_{0:\,i-1}, \beta\right) = \mathcal{N}\left(\boldsymbol{\theta}_{i-1}|\boldsymbol{\mu}_{\boldsymbol{\theta}_{i-1}}^+, \beta^{-1}\mathbf{E}_{\boldsymbol{\theta}_{i-1}}^+\right) \tag{8}$$

$$p\left(\mathbf{x}_{i-1}|\boldsymbol{\theta}_{0:\,i-1}^c, \mathbf{y}_{0:\,i-1}, \beta\right) = \mathcal{N}\left(\mathbf{x}_{i-1}|\boldsymbol{\mu}_{\mathbf{x}_{i-1}}^+, \beta^{-1}\mathbf{E}_{\mathbf{x}_{i-1}}^+\right) \tag{9}$$

It will be demonstrated later that the derived posterior PDFs $\left(\boldsymbol{\theta}_i|\boldsymbol{\theta}_{0:\,i}^c, \mathbf{y}_{0:\,i}\right)$ and $p\left(\mathbf{x}_i|\boldsymbol{\theta}_{0:\,i}^c, \mathbf{y}_{0:\,i}\right)$ for the ith time instant have the same Gaussian form as that in (8) and (9), and thus the Gaussian assumption in (8) and (9) is self-consistent. Note also that we assume the two posterior PDFs in (8) and (9) are independent of $\left\{\boldsymbol{\theta}_j\right\}_{j=0}^{i-2}$ and $\left\{\mathbf{x}_j\right\}_{j=0}^{i-2}$, respectively.

In the next section, we investigate Kalman filters to calculate the posterior PDFs of the hidden states $\mathbf{x}_i$ and parameters $\boldsymbol{\theta}_i$ conditional on a sequence of observed data. In particular, we would like to recursively estimate the posterior PDFs $p\left(\mathbf{x}_i|\mathbf{y}_{0:\,i}, \boldsymbol{\theta}_{0:\,i}^c\right)$ and $p\left(\boldsymbol{\theta}_i|\mathbf{y}_{0:\,i}, \boldsymbol{\theta}_{0:\,i}^c\right)$ considering all measurements up to the ith time instant. As stated before, there are two approaches to do so: joint estimation and dual estimation. Dual estimation is chosen here because of the benefit that the two linear-in-the-parameter equations in (2) can be utilized in two separate Kalman filters, with some information exchange between them.

3.2.1. Kalman filter for model parameter vector $\boldsymbol{\theta}_i$

Following the formulation in Appendix B, the predictive PDF $p\left(\boldsymbol{\theta}_i|\mathbf{y}_{0:\,i-1}, \boldsymbol{\theta}_{0:\,i-1}^c, \boldsymbol{\lambda}_i, \beta\right)$ for the time instant t_i is given by:

$$\begin{aligned} p\left(\boldsymbol{\theta}_i|\mathbf{y}_{0:\,i-1}, \boldsymbol{\theta}_{0:\,i-1}^c, \boldsymbol{\lambda}_i, \beta\right) &= \int p\left(\boldsymbol{\theta}_i|\boldsymbol{\theta}_{i-1}, \boldsymbol{\lambda}_i, \beta\right) p\left(\boldsymbol{\theta}_{i-1}|\mathbf{y}_{0:\,i-1}, \boldsymbol{\theta}_{0:\,i-1}^c, \beta\right) d\boldsymbol{\theta}_{i-1} \\ &= \mathcal{N}\left(\boldsymbol{\theta}_i|\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^-, \beta^{-1}\mathbf{E}_{\boldsymbol{\theta}_i}^-\right) \end{aligned} \tag{10}$$

where:

$$\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^- = \boldsymbol{\mu}_{\boldsymbol{\theta}_{i-1}}^+ \tag{11a}$$

$$\mathbf{E}_{\boldsymbol{\theta}_i}^- = \boldsymbol{\Lambda}_i + \mathbf{E}_{\boldsymbol{\theta}_{i-1}}^+ \tag{11b}$$

When the data vector $\mathbf{y}_i$ is available for the time instant t_i, the updated PDF $p\left(\boldsymbol{\theta}_i|\mathbf{y}_{0:\,i}, \boldsymbol{\theta}_{0:\,i-1}^c\right)$ is obtained from Bayes' theorem:

$$\begin{aligned} p\left(\boldsymbol{\theta}_i|\mathbf{y}_{0:\,i}, \boldsymbol{\theta}_{0:\,i-1}^c\right) &\approx p\left(\boldsymbol{\theta}_i|\mathbf{y}_{0:\,i}, \boldsymbol{\theta}_{0:\,i-1}^c, \tilde{\boldsymbol{\lambda}}_i, \tilde{\beta}\right) \\ &\propto p\left(\mathbf{y}_i|\boldsymbol{\theta}_i, \tilde{\beta}, \mathbf{y}_{0:\,i-1}\right) p\left(\boldsymbol{\theta}_i|\mathbf{y}_{0:\,i-1}, \boldsymbol{\theta}_{0:\,i-1}^c, \tilde{\boldsymbol{\lambda}}_i, \tilde{\beta}\right) \\ &= \mathcal{N}\left(\boldsymbol{\theta}_i|\tilde{\boldsymbol{\mu}}_{\boldsymbol{\theta}_i}^{+(1)}, \tilde{\beta}^{-1}\tilde{\mathbf{E}}_{\boldsymbol{\theta}_i}^{+(1)}\right) \end{aligned} \tag{12}$$

where:

$$\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^{+(1)} = \mathbf{E}_{\boldsymbol{\theta}_i}^{+(1)}\left(\mathbf{B}_i{}^T\left(\mathbf{I}_{K_i} + \mathbf{C}_i\mathbf{E}_{\mathbf{x}_i}^-\mathbf{C}_i{}^T\right)^{-1}\left(\mathbf{y}_i - \mathbf{b}_i\right) + \left(\mathbf{E}_{\boldsymbol{\theta}_i}^-\right)^{-1}\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^-\right) \tag{13a}$$

$$\mathbf{E}_{\boldsymbol{\theta}_i}^{+(1)} = \left(\left(\mathbf{E}_{\boldsymbol{\theta}_i}^{-} \right)^{-1} + \mathbf{B}_i{}^T \left(\mathbf{I}_{K_i} + \mathbf{C}_i \mathbf{E}_{\mathbf{x}_i}^{-} \mathbf{C}_i{}^T \right) \mathbf{B}_i \right)^{-1} \tag{13b}$$

$$\tilde{\boldsymbol{\mu}}_{\boldsymbol{\theta}_i}^{+(1)} = \boldsymbol{\mu}_{\boldsymbol{\theta}_i}^{+(1)} \left(\tilde{\boldsymbol{\lambda}}_i, \tilde{\beta} \right); \tilde{\mathbf{E}}_{\boldsymbol{\theta}_i}^{+(1)} = \mathbf{E}_{\boldsymbol{\theta}_i}^{+(1)} \left(\tilde{\boldsymbol{\lambda}}_i, \tilde{\beta} \right) \tag{13c}$$

and the MAP values $\left\{ \tilde{\boldsymbol{\lambda}}_i, \tilde{\beta} \right\}$ in (12) are obtained by maximizing their joint posterior PDF $p\left(\boldsymbol{\lambda}_i, \beta | \mathbf{y}_{0:\,i}, \boldsymbol{\theta}_{0:\,i-1}^{c} \right)$, giving (see Appendix B):

$$\tilde{\lambda}_{i,n} = (\mathbf{Z}_i)_{n,n} + \tilde{\beta} \left(\boldsymbol{\kappa}_i - \boldsymbol{\mu}_{\boldsymbol{\theta}_i}^{-} \right)_n^2 \tag{14}$$

$$\tilde{\beta} = a_i / b_i \tag{15}$$

where:

$$\mathbf{Z}_i = \left(\mathbf{B}_i{}^T \boldsymbol{\Gamma}_i^{-1} \mathbf{B}_i + \boldsymbol{\Lambda}_i^{-1} \right)^{-1} \tag{16a}$$

$$\boldsymbol{\kappa}_i = \mathbf{Z}_i \left(\boldsymbol{\Lambda}_i^{-1} \boldsymbol{\mu}_{\boldsymbol{\theta}_i}^{-} + \mathbf{B}_i{}^T \boldsymbol{\Gamma}_i^{-1} \left(\mathbf{y}_i - \mathbf{b}_i \right) \right) \tag{16b}$$

$$\boldsymbol{\Gamma}_i = \mathbf{B}_i \mathbf{E}_{\boldsymbol{\theta}_{i-1}}^{+} \mathbf{B}_i{}^T + \mathbf{I}_{K_i} + \mathbf{C}_i \mathbf{E}_{\mathbf{x}_i}^{-} \mathbf{C}_i{}^T \tag{16c}$$

$$a_i = a_{i-1} + K_i / 2 \tag{16d}$$

$$b_i = b_{i-1} + \frac{1}{2} \left[\left(\mathbf{y}_i - \mathbf{B}_i \boldsymbol{\mu}_{\boldsymbol{\theta}_i}^{-} - \mathbf{b}_i \right)^T \left(\mathbf{I}_{K_i} + \mathbf{B}_i \mathbf{E}_{\boldsymbol{\theta}_i}^{-} \mathbf{B}_i{}^T + \mathbf{C}_i \mathbf{E}_{\mathbf{x}_i}^{-} \mathbf{C}_i{}^T \right)^{-1} \left(\mathbf{y}_i - \mathbf{B}_i \boldsymbol{\mu}_{\boldsymbol{\theta}_i}^{-} - \mathbf{b}_i \right) \right] \tag{16e}$$

Based on the next pseudo-data $\boldsymbol{\theta}_i^c$, the posterior PDF for $\boldsymbol{\theta}_i$ is updated based on Bayes' theorem:

$$\begin{aligned} & p\left(\boldsymbol{\theta}_i | \mathbf{y}_{0:\,i}, \boldsymbol{\theta}_{0:\,i}^{c} \right) \approx p\left(\boldsymbol{\theta}_i | \mathbf{y}_{0:\,i}, \boldsymbol{\theta}_{0:\,i}^{c}, \tilde{\boldsymbol{\alpha}}_i \right) \\ & \propto p\left(\boldsymbol{\theta}_i^c | \boldsymbol{\theta}_i, \mathbf{y}_{0:\,i}, \boldsymbol{\theta}_{0:\,i-1}^{c}, \tilde{\boldsymbol{\alpha}}_i \right) p\left(\boldsymbol{\theta}_i | \mathbf{y}_{0:\,i}, \boldsymbol{\theta}_{0:\,i-1}^{c} \right) \\ & = \mathcal{N}\left(\boldsymbol{\theta}_i | \tilde{\boldsymbol{\mu}}_{\boldsymbol{\theta}_i}^{+}, \tilde{\beta}^{-1} \tilde{\mathbf{E}}_{\boldsymbol{\theta}_i}^{+} \right) \end{aligned} \tag{17}$$

where:

$$\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^{+} = \mathbf{E}_{\boldsymbol{\theta}_i}^{+} \left\{ \left(\mathbf{E}_{\boldsymbol{\theta}_i}^{+(1)} \right)^{-1} \boldsymbol{\mu}_{\boldsymbol{\theta}_i}^{+(1)} + \mathbf{A}_i^{-1} \boldsymbol{\theta}_i^c \right\} \tag{18a}$$

$$\mathbf{E}_{\boldsymbol{\theta}_i}^{+} = \left(\left(\mathbf{E}_{\boldsymbol{\theta}_i}^{+(1)} \right)^{-1} + \mathbf{A}_i^{-1} \right)^{-1} \tag{18b}$$

$$\tilde{\boldsymbol{\mu}}_{\boldsymbol{\theta}_i}^{+} = \boldsymbol{\mu}_{\boldsymbol{\theta}_i}^{+} \left(\tilde{\boldsymbol{\lambda}}_i, \tilde{\beta}, \tilde{\alpha}_i \right); \tilde{\mathbf{E}}_{\boldsymbol{\theta}_i}^{+} = \mathbf{E}_{\boldsymbol{\theta}_i}^{+} \left(\tilde{\boldsymbol{\lambda}}_i, \tilde{\beta}, \tilde{\alpha}_i \right)$$

The MAP values of the hyper-parameters $\boldsymbol{\alpha}_i$ are given by:

$$\tilde{\alpha}_{i,n} = \left(\mathbf{E}_{\boldsymbol{\theta}_i}^{+} \right)_{n,n} + \beta \left(\boldsymbol{\theta}_i^c - \boldsymbol{\mu}_{\boldsymbol{\theta}_i}^{+} \right)_n^2 \tag{19}$$

Since the MAP estimates $\tilde{\boldsymbol{\lambda}}_i, \tilde{\beta}$ and $\tilde{\boldsymbol{\alpha}}_i$ are interrelated, an iterative method that cycles over (13a)–(13c), (14), (15), (18a)–(18b) and (19) until convergence is required to compute them.

3.2.2. *Kalman filter for system state vector* $\mathbf{x}_i$

Following the formulation in Appendix A, the predictive PDF:

$$\begin{aligned} p\left(\mathbf{x}_i | \mathbf{y}_{0:\,i-1}, q_i, \beta \right) &= \iint p\left(\mathbf{x}_i | \mathbf{x}_{i-1}, \boldsymbol{\theta}_{i-1}, q_i, \beta \right) p\left(\mathbf{x}_{i-1} | \mathbf{y}_{0:\,i-1} \right) d\mathbf{x}_{i-1} . p\left(\boldsymbol{\theta}_{i-1} | \mathbf{y}_{0:\,i-1} \right) d\boldsymbol{\theta}_{i-1} \\ &= \mathcal{N}\left(\mathbf{x}_i | \boldsymbol{\mu}_{\mathbf{x}_i}^{-}, \boldsymbol{\Sigma}_{\mathbf{x}_i}^{-} \right) \end{aligned} \tag{20}$$

where:

$$\boldsymbol{\mu}_{\mathbf{x}_i}^{-} = \mathbf{P}_{i-1} \left(\boldsymbol{\mu}_{\boldsymbol{\theta}_{i-1}}^{+} \right) \boldsymbol{\mu}_{\mathbf{x}_{i-1}}^{+} + \mathbf{p}_{i-1} \left(\boldsymbol{\mu}_{\boldsymbol{\theta}_{i-1}}^{+} \right) \tag{21a}$$

$$\mathbf{E}_{\mathbf{x}_i}^{-} = q_i \mathbf{I}_{M_i} + \mathbf{P}_{i-1}\left(\boldsymbol{\mu}_{\boldsymbol{\theta}_{i-1}}^{+}\right)\mathbf{E}_{\mathbf{x}_{i-1}}^{+}\mathbf{P}_{i-1}^{T}\left(\boldsymbol{\mu}_{\boldsymbol{\theta}_{i-1}}^{+}\right) + \mathbf{L}_{i-1}\left(\boldsymbol{\mu}_{\boldsymbol{\theta}_{i-1}}^{+}\right)\mathbf{E}_{\boldsymbol{\theta}_{i-1}}^{+}\mathbf{L}_{i-1}^{T}\left(\boldsymbol{\mu}_{\boldsymbol{\theta}_{i-1}}^{+}\right) \tag{21b}$$

$$\mathbf{L}_{i-1} = \frac{\partial f_{i-1}(\boldsymbol{\mu}_{\mathbf{x}_{i-1}}^{+}, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}} + \frac{\partial f_{i-1}(\boldsymbol{\mu}_{\mathbf{x}_{i-1}}^{+}, \boldsymbol{\theta})}{\partial \boldsymbol{\mu}_{\mathbf{x}_{i-1}}^{+}} \cdot \frac{\partial \boldsymbol{\mu}_{\mathbf{x}_{i-1}}^{+}}{\partial \boldsymbol{\theta}} = \frac{\partial f_{i-1}(\boldsymbol{\mu}_{\mathbf{x}_{i-1}}^{+}, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}} + \mathbf{P}_{i-1} \cdot \frac{\partial \boldsymbol{\mu}_{\mathbf{x}_{i-1}}^{+}}{\partial \boldsymbol{\theta}} \tag{21c}$$

When new data vector $\mathbf{y}_i$ is available, the conditional posterior PDF $p\left(\mathbf{x}_i|\boldsymbol{\theta}_i, \boldsymbol{\theta}_{0:\,i}^{c}, \mathbf{y}_{0:\,i}\right)$ is computed according to the procedure in Appendix B:

$$p\left(\mathbf{x}_i|\mathbf{y}_{0:\,i}, \boldsymbol{\theta}_{0:\,i}^{c}\right) = p\left(\mathbf{x}_i|\mathbf{y}_{0:\,i}, \boldsymbol{\theta}_{0:\,i-1}^{c}\right) \propto p\left(\mathbf{y}_i|\mathbf{x}_i, \tilde{\beta}\right) p\left(\mathbf{x}_i|\mathbf{y}_{0:\,i-1}, \boldsymbol{\theta}_{0:\,i-1}^{c}\right) = \mathcal{N}\left(\mathbf{x}_i|\tilde{\boldsymbol{\mu}}_{\mathbf{x}_i}^{+}, \beta^{-1}\tilde{\mathbf{E}}_{\mathbf{x}_i}^{+}\right) \tag{22}$$

where:

$$\left\{\tilde{q}_i, \tilde{\beta}\right\} = \operatorname{argmax} p\left(q_i, \beta|\boldsymbol{\theta}_{0:\,i-1}^{c}, \mathbf{y}_{0:\,i}\right) \tag{23a}$$

$$\boldsymbol{\mu}_{\mathbf{x}_i}^{+} = \mathbf{E}_{\mathbf{x}_i}^{+}\left(\left(\mathbf{E}_{\mathbf{x}_i}^{-}\right)^{-1}\boldsymbol{\mu}_{\mathbf{x}_i}^{-} + \mathbf{C}_i{}^{T}\left(\mathbf{I}_{K_i} + \mathbf{B}_i\mathbf{E}_{\boldsymbol{\theta}_i}^{-}\mathbf{B}_i{}^{T}\right)^{-1}\left(\mathbf{y}_i - \mathbf{c}_i\right)\right) \tag{23b}$$

$$\mathbf{E}_{\mathbf{x}_i}^{+} = \left(\left(\mathbf{E}_{\mathbf{x}_i}^{-}\right)^{-1} + \mathbf{C}_i{}^{T}\left(\mathbf{I}_{K_i} + \mathbf{B}_i\mathbf{E}_{\boldsymbol{\theta}_i}^{-}\mathbf{B}_i{}^{T}\right)\mathbf{C}_i\right)^{-1} \tag{23c}$$

$$\tilde{\boldsymbol{\mu}}_{\mathbf{x}_i}^{-} = \boldsymbol{\mu}_{\mathbf{x}_i}^{-}\left(\tilde{q}_i\right); \tilde{\mathbf{E}}_{\mathbf{x}_i}^{-} = \mathbf{E}_{\mathbf{x}_i}^{-}\left(\tilde{q}_i\right); \tilde{\boldsymbol{\mu}}_{\mathbf{x}_i}^{+} = \boldsymbol{\mu}_{\mathbf{x}_i}^{+}\left(\tilde{q}_i, \tilde{\beta}\right); \tilde{\mathbf{E}}_{\mathbf{x}_i}^{+} = \mathbf{E}_{\mathbf{x}_i}^{+}\left(\tilde{q}_i, \tilde{\beta}\right)$$

The MAP values $\left\{\tilde{q}_i, \tilde{\beta}\right\}$ in (23a) are computed by maximizing their conditional posterior PDF $p\left(q_i, \beta|\boldsymbol{\theta}_{0:\,i-1}^{c}, \mathbf{y}_{0:\,i}\right) \propto p\left(\mathbf{y}_i|\boldsymbol{\theta}_{0:\,i-1}^{c}, \mathbf{y}_{0:\,i-1}, q_i, \beta\right) p\left(q_i, \beta|\boldsymbol{\theta}_{0:\,i-1}^{c}, \mathbf{y}_{0:\,i-1}\right)$, giving:

$$\tilde{q}_i = \left(\operatorname{tr}\left(\boldsymbol{\Pi}_i\right) + \tilde{\beta}\left\|\boldsymbol{\eta}_i - \boldsymbol{\mu}_{\mathbf{x}_i}^{-}\right\|^2\right)/M_i \tag{24}$$

$$\tilde{\beta} = a_i/b_i \tag{25}$$

where:

$$\boldsymbol{\Pi}_i = \left(\mathbf{C}_i{}^{T}\mathbf{T}_i^{-1}\mathbf{C}_i + q_i^{-1}\mathbf{I}_{M_i}\right)^{-1} \tag{26a}$$

$$\boldsymbol{\eta}_i = \boldsymbol{\Pi}_i\left(\mathbf{C}_i^{T}\mathbf{T}_i^{-1}\left(\mathbf{y}_i - \mathbf{c}_i\right) + q_i^{-1}\boldsymbol{\mu}_{\mathbf{x}_i}^{-}\right) \tag{26b}$$

$$\mathbf{T}_i = \mathbf{C}_i\mathbf{P}_i\mathbf{E}_{\mathbf{x}_{i-1}}^{+}\mathbf{P}_i^{T}\mathbf{C}_i{}^{T} + \mathbf{I}_{K_i} + \mathbf{B}_i\mathbf{E}_{\boldsymbol{\theta}_i}^{-}\mathbf{B}_i{}^{T} \tag{26c}$$

$$a_i = a_{i-1} + K_i/2 \tag{26d}$$

$$b_i = b_{i-1} + \left(\mathbf{y}_i - \mathbf{C}_i\boldsymbol{\mu}_{\mathbf{x}_i}^{-} - \mathbf{c}_i\right)^{T}\left(\mathbf{I}_{K_i} + \mathbf{C}_i\mathbf{E}_{\mathbf{x}_i}^{-}\mathbf{C}_i{}^{T} + \mathbf{B}_i\mathbf{E}_{\boldsymbol{\theta}_i}^{-}\mathbf{B}_i{}^{T}\right)^{-1}\left(\mathbf{y}_i - \mathbf{C}_i\boldsymbol{\mu}_{\mathbf{x}_i}^{-} - \mathbf{c}_i\right)/2 \tag{26e}$$

Since $\tilde{q}_i$ and $\tilde{\beta}$ are interrelated, an iterative method that cycles over (23b)–(23c), (24) and (25), until convergence is required to compute their MAP values.

Remark 2. In (16e) and (26e), $\mathbf{B}_i\left(\boldsymbol{\mu}_{\mathbf{x}_i}^{-}\right)\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^{-} - \mathbf{b}_i\left(\boldsymbol{\mu}_{\mathbf{x}_i}^{-}\right) = \mathbf{C}_i\left(\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^{-}\right)\boldsymbol{\mu}_{\mathbf{x}_i}^{-} - \mathbf{c}_i\left(\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^{-}\right)$ and so the MAP estimates of β in (15) and (25) for the separate Kalman filters for $\boldsymbol{\theta}_i$ and $\mathbf{x}_i$ are the same. It is also seen that the posterior means $\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^{+}$ in (18a) and $\boldsymbol{\mu}_{\mathbf{x}_i}^{+}$ in (23b) and posterior covariance matrices $\mathbf{E}_{\boldsymbol{\theta}_i}^{+}$ in (18b) and $\mathbf{E}_{\mathbf{x}_i}^{+}$ in (23c) are interrelated, because of the correlation between the state and parameter vectors. Therefore, an iterative procedure that cycles between the two Kalman filters until convergence is required to compute the posterior means $\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^{+}$ and $\boldsymbol{\mu}_{\mathbf{x}_i}^{+}$, and the posterior scaled covariance matrices $\mathbf{E}_{\boldsymbol{\theta}_i}^{+}$ and $\mathbf{E}_{\mathbf{x}_i}^{+}$.

Remark 3. If the model parameter $\boldsymbol{\theta}_i$ is known and we choose $\mathbf{x}_i$ solely as the state vector, the Bayesian sequential state updating for $\mathbf{x}_i$ in the previous section is identical to the traditional Kalman formulation; Similarly, if we choose $\boldsymbol{\theta}_i$ solely as the state vector and set all $\alpha_{i,n} \to \infty, n = 1, \ldots, N$, the sequential updating results for $\boldsymbol{\theta}_i$ are also equal to the traditional Kalman filter formulation.

3.3. Pseudo-codes for the sparseness inducing dual Kalman filter algorithm

To summarize the sparseness inducing dual Kalman filter for each single time instant t_i, Algorithm 1 below is provided.

Algorithm 1: Sparseness-inducing dual Kalman filter for a single time instant t_i

Sequential inputs:

▷ The measurements $\mathbf{y}_i$ for the time instant t_i.

▷ The calibration value $\boldsymbol{\theta}_i^c$ for $\boldsymbol{\theta}_i$

▷ Posterior means $\boldsymbol{\mu}_{\mathbf{x}_{i-1}}^+$ and $\boldsymbol{\mu}_{\boldsymbol{\theta}_{i-1}}^+$ and scaled covariance matrices $\mathbf{E}_{\mathbf{x}_{i-1}}^+$ and $\mathbf{E}_{\boldsymbol{\theta}_{i-1}}^+$ conditional on the measurements up to the $(i-1)^{th}$ time step, and set the initial value of $\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^+ = \boldsymbol{\mu}_{\boldsymbol{\theta}_{i-1}}^+$

▷ Matrix $\mathbf{P}_{i-1}(\boldsymbol{\mu}_{\boldsymbol{\theta}_{i-1}}^+)$ and vector $\mathbf{p}_{i-1}(\boldsymbol{\mu}_{\boldsymbol{\theta}_{i-1}}^+)$

▷ Initial values of the MAP parameters of $\tilde{\boldsymbol{\lambda}}_i, \tilde{\boldsymbol{\alpha}}_i, \tilde{q}_i, \tilde{\beta}$

Algorithm:

1. **While** convergence criterion on $\left[\left(\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^+\right)^T, \left(\boldsymbol{\mu}_{\mathbf{x}_i}^+\right)^T\right]^T$ is not met (outer loop)
2. **While** convergence criterion on $\boldsymbol{\mu}_{\mathbf{x}_i}^+$ is not met (the 1st inner loop)
3. Compute $\mathbf{C}_i\left(\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^+\right)$ and $\mathbf{c}_i\left(\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^+\right)$
4. Compute $\boldsymbol{\mu}_{\mathbf{x}_i}^-$ and $\mathbf{E}_{\mathbf{x}_i}^-$ given $\tilde{q}_i$ using (21a) and (21b)
5. Compute $\boldsymbol{\mu}_{\mathbf{x}_i}^+$ and $\mathbf{E}_{\mathbf{x}_i}^+$ given $\tilde{q}_i$ and $\tilde{\beta}$ using (23b) and (23c)
6. Compute the MAP value $\tilde{q}_i$ using (24)
7. Compute the MAP value $\tilde{\beta}$ using (25)
8. End **While**
9. **While** convergence criterion on $\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^+$ is not met (the 2nd inner loop)
10. Compute $\mathbf{B}_i\left(\boldsymbol{\mu}_{\mathbf{x}_i}^+\right)$ and $\mathbf{b}_i\left(\boldsymbol{\mu}_{\mathbf{x}_i}^+\right)$
11. Compute $\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^-$ and $\mathbf{E}_{\boldsymbol{\theta}_i}^-$ given $\tilde{\boldsymbol{\lambda}}_i$ using (11) and (11b)
12. Compute $\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^{+(1)}$ and $\mathbf{E}_{\boldsymbol{\theta}_i}^{+(1)}$ given $\tilde{\beta}$ and $\tilde{\boldsymbol{\lambda}}_i$ from (13a) and (13b)
13. Compute the MAP value $\tilde{\boldsymbol{\lambda}}_i$ using (14)
14. Compute the MAP value $\tilde{\beta}$ using (15)
15. Compute $\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^+$ and $\mathbf{E}_{\boldsymbol{\theta}_i}^+$ given $\tilde{\boldsymbol{\alpha}}_i$, $\tilde{r}_i$ and $\tilde{\boldsymbol{\lambda}}_i$ from (18a) and (18b)
16. Compute the MAP value $\tilde{\boldsymbol{\alpha}}_i$ using (19)
17. End **While**
18. End **While**

OUTPUTS:

▷ Posterior means $\boldsymbol{\mu}_{\mathbf{x}_i}^+$ and $\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^+$ and posterior covariance matrices $\boldsymbol{\Sigma}_{\mathbf{x}_i}^+ = \tilde{\beta}^{-1}\,\mathbf{E}_{\mathbf{x}_i}^+$ and $\boldsymbol{\Sigma}_{\boldsymbol{\theta}_i}^+ = \tilde{\beta}^{-1}\mathbf{E}_{\boldsymbol{\theta}_i}^+$ conditional on the measurements up to the ith time step

We see that the process essentially comprises two Kalman filters running in parallel with some information exchange between them. The first and second inner loops adapt the system state vector $\mathbf{x}_i$ and model parameter vector $\boldsymbol{\theta}_i$, respectively, which are terminated when the changes in the posterior mean values are sufficiently small, e.g. $\left\|\left(\boldsymbol{\mu}_{\mathbf{x}_i}^{+}\right)^{[l+1]}-\left(\boldsymbol{\mu}_{\mathbf{x}_i}^{+}\right)^{[l]}\right\|_2^2 / \left\|\left(\boldsymbol{\mu}_{\mathbf{x}_i}^{+}\right)^{[l]}\right\|_2^2 < \epsilon$ and $\left\|\left(\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^{+}\right)^{[l+1]}-\left(\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^{+}\right)^{[l]}\right\|_2^2 / \left\|\left(\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^{+}\right)^{[l]}\right\|_2^2 < \epsilon$, where l is the iteration number for an inner loop and ϵ is a specified threshold (we take $\epsilon = 0.01$ in the examples later to avoid an excessive number of iterations). A similar convergence criterion is also employed for $\left[\left(\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^{+}\right)^T, \left(\boldsymbol{\mu}_{\mathbf{x}_i}^{+}\right)^T\right]^T$ for the outer loop, i.e., $\left\|\left(\left[\left(\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^{+}\right)^T, \left(\boldsymbol{\mu}_{\mathbf{x}_i}^{+}\right)^T\right]^T\right)^{[m+1]} - \left(\left[\left(\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^{+}\right)^T, \left(\boldsymbol{\mu}_{\mathbf{x}_i}^{+}\right)^T\right]^T\right)^{[m]}\right\|_2^2 / \left\|\left(\left[\left(\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^{+}\right)^T, \left(\boldsymbol{\mu}_{\mathbf{x}_i}^{+}\right)^T\right]^T\right)^{[m]}\right\|_2^2 < \epsilon$, where m is the iteration number for the outer loop.

4. Illustrative examples

4.1. Structural state–space model

Consider a linear structural dynamics problem formulated by the continuous time second-order differential equation:

$$\mathbf{M}_s\ddot{\mathbf{z}}(t) + \mathbf{C}_s\dot{\mathbf{z}}(t) + \mathbf{K}_s\mathbf{z}(t) = \mathbf{f}(t) = \mathbf{S}_p\mathbf{u}(t) \tag{27}$$

where $\mathbf{z}(t) \in \mathbb{R}^{N_d}$ denotes the displacement vector, $\mathbf{f}(t) \in \mathbb{R}^{N_d}$ is the known excitation force, which is presented as a superposition of time histories $\mathbf{u}(t) \in \mathbb{R}^{N_f}$ that are influencing some DOFs of the structure as indicated via the force distribution matrix $\mathbf{S}_p \in \mathbb{R}^{N_d \times N_f}$, and $\mathbf{M}_s$, $\mathbf{C}_s$ and $\mathbf{K}_s$ are the mass, damping and stiffness matrix of the structural system, respectively. The structural stiffness matrix $\mathbf{K}_s = \mathbf{K}_0 + \sum_{j=1}^{N_w} w_j\mathbf{K}_j$ is uncertain, where the substructure stiffness matrix contribution $\mathbf{K}_j$ are chosen a priori (e.g. by using a finite model). The structural stiffness scaling parameter $\mathbf{w}(t) \in \mathbb{R}^{N_w}$ is the structural model parameter and it will be tracked as a function of time t. The Rayleigh damping model is adopted, so the damping matrix is given by $\mathbf{C}_s = a\mathbf{M}_s + b\mathbf{K}_s$, where a and b are chosen a priori. The mass matrix $\mathbf{M}_s$ can be chosen a priori based on structural drawings.

The continuous equations in Eq. (27) can be converted to a state–space equation, where the displacement vector $\mathbf{z}(t)$ and the velocity vector $\dot{\mathbf{z}}(t)$ are employed to define the usual state vector $\mathbf{x}(t) = \left[\mathbf{z}(t)^T, \dot{\mathbf{z}}(t)^T\right]^T \in \mathbb{R}^{2N_d}$:

$$\dot{\mathbf{x}}(t) = \begin{bmatrix}\dot{\mathbf{z}}(t)\\ \ddot{\mathbf{z}}(t)\end{bmatrix} = \mathbf{P}_c(\mathbf{w}(t))\begin{bmatrix}\mathbf{z}(t)\\ \dot{\mathbf{z}}(t)\end{bmatrix} + \mathbf{H}_c\mathbf{u}(t) + \boldsymbol{\varepsilon}(t) = \mathbf{P}_c(\mathbf{w}(t))\,\mathbf{x}(t) + \mathbf{H}_c\mathbf{u}(t) + \boldsymbol{\varepsilon}(t) \tag{28}$$

where:

$$\mathbf{P}_c(\mathbf{w}(t)) = \begin{bmatrix}\mathbf{O}_{N_d\times N_d} & \mathbf{I}_{N_d\times N_d}\\ -\mathbf{M}_s^{-1}\mathbf{K}_s(\mathbf{w}) & -\mathbf{M}_s^{-1}\mathbf{C}_s(\mathbf{w})\end{bmatrix}_{2N_d\times 2N_d} \tag{29a}$$

$$\mathbf{H}_c = \begin{bmatrix}\mathbf{O}_{N_d\times N_F}\\ -\mathbf{M}_s^{-1}\mathbf{S}_p\end{bmatrix}_{2N_d\times N_F} \tag{29b}$$

and we have introduced a state–space model prediction error $\boldsymbol{\varepsilon}(t)$.

It is assumed that the accelerations of the structural system are measured at N_m DOFs with sampling interval Δt and this output $\mathbf{y}(t) \in \mathbb{R}^{N_m}$ at time t is modeled by:

$$\mathbf{y}(t) = \mathbf{C}_c(\mathbf{w}(t))\begin{bmatrix}\mathbf{z}(t)\\ \dot{\mathbf{z}}(t)\end{bmatrix} + \mathbf{G}_c\mathbf{u}(t) + \mathbf{e}(t) = \mathbf{C}_c(\mathbf{w}(t))\mathbf{x}(t) + \mathbf{G}_c\mathbf{u}(t) + \mathbf{e}(t) \tag{30}$$

where we have introduced a combined observation model prediction error and measurement error $\mathbf{e}(t)$ and the observation matrix $\mathbf{C}_c(\mathbf{w}(t))$ and direct transmission matrix $\mathbf{G}_c$ are given by:

$$\begin{aligned}\mathbf{C}_c\left(\mathbf{w}\left(t\right)\right) &= \left[-\mathbf{S}_a\mathbf{M}_s^{-1}\mathbf{K}_s\left(\mathbf{w}\left(t\right)\right) - \mathbf{S}_a\mathbf{M}_s^{-1}\mathbf{C}_s\right]_{N_m\times N_d}\\ &= \left[-\mathbf{S}_a\mathbf{M}_s^{-1}\mathbf{K}_s\left(\mathbf{w}\left(t\right)\right) - \mathbf{S}_a\left(a\mathbf{I}_{N_d} + b\mathbf{M}_s^{-1}\mathbf{K}_s\left(\mathbf{w}\left(t\right)\right)\right)\right]_{N_m\times N_d}\end{aligned} \tag{31a}$$

$$\mathbf{G}_c = \mathbf{S}_a\mathbf{M}_s^{-1}\mathbf{S}_p \tag{31b}$$

where $\mathbf{S}_a \in \mathbb{R}^{N_m\times N_d}$ is the selection matrix of appropriate dimension for the accelerations. Note that the measurement $\mathbf{y}\left(t\right)$ also can be expressed linearly with respect to the stiffness scaling parameter vector $\mathbf{w}$:

$$\begin{aligned}\mathbf{y}\left(t\right) &= -\mathbf{S}_a\mathbf{M}_s^{-1}\mathbf{K}_s\left(\mathbf{w}\left(t\right)\right)\mathbf{z}\left(t\right) - \mathbf{S}_a\left(a\mathbf{I}_{N_d} + b\mathbf{M}_s^{-1}\mathbf{K}_s\left(\mathbf{w}\left(t\right)\right)\right)\dot{\mathbf{z}}\left(t\right) + \mathbf{G}_c\mathbf{u}\left(t\right) + \mathbf{e}\left(t\right)\\ &= -\mathbf{S}_a\mathbf{M}_s^{-1}\boldsymbol{L}\left(\mathbf{z}\left(t\right)\right)\mathbf{w}\left(t\right) - b\mathbf{S}_a\mathbf{M}_s^{-1}\boldsymbol{J}\left(\dot{\mathbf{z}}\left(t\right)\right)\mathbf{w}\left(t\right) - a\mathbf{S}_a\dot{\mathbf{z}}\left(t\right) - \mathbf{S}_a\mathbf{M}_s^{-1}\mathbf{K}_0\left(\mathbf{z}\left(t\right) + b\dot{\mathbf{z}}\left(t\right)\right)\\ &\quad + \mathbf{G}_c\mathbf{u}\left(t\right) + \mathbf{e}\left(t\right)\end{aligned} \tag{32}$$

since:

$$\begin{aligned}\mathbf{K}_s\left(\mathbf{w}\left(t\right)\right)\mathbf{z}\left(t\right) &= \sum_{j=1}^{N_\theta} w_j\mathbf{K}_j\mathbf{z}\left(t\right) + \mathbf{K}_0\mathbf{z}\left(t\right) = \boldsymbol{L}\left(\mathbf{z}\left(t\right)\right)\mathbf{w}\left(t\right) + \mathbf{K}_0\mathbf{z}\left(t\right)\\ \mathbf{K}_s\left(\mathbf{w}\left(t\right)\right)\dot{\mathbf{z}}\left(t\right) &= \sum_{j=1}^{N_\theta} w_j\mathbf{K}_j\dot{\mathbf{z}}\left(t\right) + \mathbf{K}_0\dot{\mathbf{z}}\left(t\right) = \boldsymbol{J}\left(\dot{\mathbf{z}}\left(t\right)\right)\boldsymbol{\theta}\left(t\right) + \mathbf{K}_0\dot{\mathbf{z}}\left(t\right)\end{aligned} \tag{33a}$$

where:

$$\begin{aligned}\boldsymbol{L}\left(\mathbf{z}\left(t\right)\right) &= \left[\mathbf{K}_1\mathbf{z}\left(t\right) \quad \cdots \quad \mathbf{K}_{N_\mathbf{w}}\mathbf{z}\left(t\right)\right]_{N_d\times N_w}\\ \boldsymbol{J}\left(\dot{\mathbf{z}}\left(t\right)\right) &= \left[\mathbf{K}_1\dot{\mathbf{z}}\left(t\right) \quad \cdots \quad \mathbf{K}_{N_w}\dot{\mathbf{z}}\left(t\right)\right]_{N_d\times N_w}\end{aligned} \tag{33b}$$

Finally, the bilinear discrete-time state space equation for the system state vector $\mathbf{x}_i = \begin{bmatrix}\mathbf{z}_i\\ \dot{\mathbf{z}}_i\end{bmatrix} = \begin{bmatrix}\mathbf{z}\left(i\,\Delta t\right)\\ \dot{\mathbf{z}}\left(i\,\Delta t\right)\end{bmatrix}$ and structural stiffness scaling parameter vector $\mathbf{w}_i = \mathbf{w}\left(i\,\Delta t\right)$ can be obtained, where the combined measurement and output prediction error $\mathbf{e}_i$ and state prediction error $\boldsymbol{\varepsilon}_i$ are now for each discrete time t_i:

$$\begin{aligned}\mathbf{y}_i &= \mathbf{C}_i\left(\mathbf{w}_i\right)\mathbf{x}_i + \mathbf{G}_i\mathbf{u}_i + \mathbf{e}_i = \mathbf{B}_i\left(\mathbf{x}_i\right)\mathbf{w}_i + \mathbf{b}_i\left(\mathbf{x}_i\right) + \mathbf{G}_i\mathbf{u}_i + \mathbf{e}_i\\ \mathbf{x}_i &= \mathbf{P}_{i-1}\left(\mathbf{w}_{i-1}\right)\mathbf{x}_{i-1} + \mathbf{H}_{i-1}\left(\mathbf{w}_{i-1}\right)\mathbf{u}_{i-1} + \boldsymbol{\varepsilon}_i\\ \mathbf{w}_i &= \mathbf{w}_{i-1} + \boldsymbol{\delta}_i \quad i\in\mathbb{Z}^+\end{aligned} \tag{34}$$

where:

$$\begin{aligned}&\mathbf{C}_i = \mathbf{C}_c\left(\mathbf{w}_i\right), \mathbf{G}_i = \mathbf{G}_c\left(\mathbf{w}_i\right),\\ &\mathbf{B}_i\left(\mathbf{x}_i\right) = -\mathbf{S}_a\mathbf{M}_s^{-1}\boldsymbol{L}_i - b\mathbf{S}_a\mathbf{M}_s^{-1}\boldsymbol{J}_i, \mathbf{b}_i\left(\mathbf{x}_i\right) = -a\mathbf{S}_a\dot{\mathbf{z}}_i - \mathbf{S}_a\mathbf{M}_s^{-1}\mathbf{K}_0\left(\mathbf{z}_i + b\dot{\mathbf{z}}_i\right),\\ &\boldsymbol{L}_i = \boldsymbol{L}\left(\mathbf{z}\left(t\right)\right), \boldsymbol{J}_i = \boldsymbol{J}\left(\dot{\mathbf{z}}\left(t\right)\right)\\ &\mathbf{P}_i = \exp\left(\mathbf{P}_c\left(\mathbf{w}_i\right)\Delta t\right), \mathbf{H}_i = \mathbf{P}_c^{-1}\left(\mathbf{w}_i\right)\left(\mathbf{P}_i\left(\mathbf{w}_i\right) - \mathbf{I}_{2N_d\times 2N_d}\right)\mathbf{H}_c\end{aligned} \tag{35}$$

In the next subsections, the structural state–space model defined in this subsection is used for illustrative examples of tracking external loads and temporal stiffness changes of a 20-DOF structural model.

4.2. Load identification

By using Algorithm 1, estimation of the state and load vectors is carried out by using simulated data for a 20-story shear frame structure where single and double-point impact loads, as well as seismic wave excitation, are applied to the shear frame. The external structural load $\mathbf{u}\left(t\right)\in\mathbb{R}^{N_f}$ is the model parameter vector to be tracked at each time instant.

The 20-story shear frame model is shown in Fig. 1. The basic information of the structural model is chosen as: the mass of each floor is 7.50×10^5 kg; the inter-story stiffness is 10^6 KN/m, giving a fundamental modal frequency of 0.4452 Hz; and for the Rayleigh damping matrix model $\mathbf{C}_s = a\mathbf{M}_s + b\mathbf{K}_s$, the damping coefficients are $a = 0.0839$ s^{-1} and $b = 0.0036$ s. The performance of our proposed algorithm for estimating different types

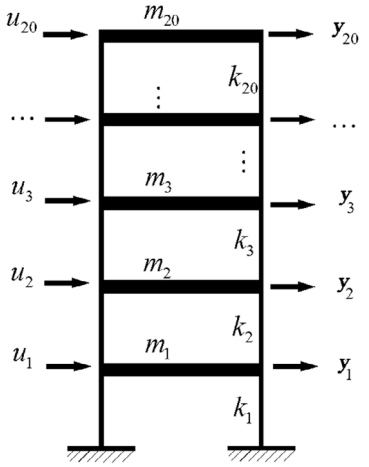


Fig. 1. 20-story shear frame model.

of loads is investigated using incomplete noisy observations of structural response from ten accelerometers, one on each of the 3rd, 5th, 7th, 8th, 10th, 13th, 15th, 17th, 18th and 20th floors (i.e. $N_m = 10$). The sensor sampling frequency is 200 Hz and discrete-time Gaussian white noise with standard deviation equal to 10% of root mean square (RMS) of the roof acceleration is added to the acceleration data to simulate measurement error.

For the load identification examples, the locations of loads are unknown and the load vector $\mathbf{u}_i \in \mathbb{R}^{N_d}$ at each time instant t_i is taken as the model parameters $\boldsymbol{\theta}_i$ in Eqs. (1)–(2) in Section 2. For the impact loads that are applied, the model load vector $\mathbf{u}_i$ is assumed to be sparse, so the calibration vector $\boldsymbol{\theta}_i^c = \mathbf{0}_{N_d}$ in Eq. (6).

4.2.1. Single-point impact load identification

In the example of single-point impact load identification, the total monitoring time is 5 s and at the time instant of 2 s, a triangular impact load of duration 0.2 s is applied to the 20th floor of the structure and it reaches a peak value of 10^3kN and then falls to zero at the time instants of 2.1 s and 2.2 s, respectively.

In Fig. 2, the estimated results for possible impact loads at the 15th, 18th and 20th floors are shown. In these figures, the solid and dashed lines represent the estimated mean and actual values respectively, and the dotted lines represent the plus and minus three standard deviation confidence intervals ($\pm 3\sigma$), corresponding to a probability of 99.7%. The same format of the confidence intervals will also be applied to later figures. It is seen that based on the accelerometer data the proposed dual Kalman filter algorithm accurately estimates the location and magnitude of the impact load acting on the 20th floor. The actual and estimated peak values of the impact load are 1000 and 972 kN, respectively, which agree to within 3%. For the 15th and 18th floors where there was no load acting, the 99.7% confidence intervals of the estimated load are within the bound of [−0.2, 0.2] kN, corresponding to 0.02% of the actual peak load on the 20th floor. Fig. 3 presents the estimated time-histories of the state vectors $\mathbf{x}(t) = \left[\mathbf{z}(t)^T, \dot{\mathbf{z}}(t)^T\right]^T$. The 99.7% confidence intervals are narrow but because of the sparseness constraints incorporated in the dual Kalman filter algorithm, they still cover the actual displacement and velocity values for the entire accelerometer monitoring duration.

In Fig. 4, the MAP values of the measurement noise variance $\tilde{\beta}^{-1}$ are compared with the actual variance of the measurement noise, which is $9.39 \times 10^{-5}\mathrm{I}_{\mathrm{N_m}}$. It is seen that the values of $\tilde{\beta}^{-1}$ gradually converge to the actual variance of the measurement noise with increasing time. In Fig. 5, the MAP values of the hyper-parameters $\lambda_{i,15}$, $\lambda_{i,20}$, $\alpha_{i,15}$ and $\alpha_{i,20}$ are also shown, where $\lambda_{i,j}$ and $\alpha_{i,\mathrm{j}}$ are the previously-defined hyper-parameters proportional to the jth component of variance for the sparse changes $\boldsymbol{\delta}_i = \mathbf{u}_i - \mathbf{u}_{i-1}$ and the sparse vector $\mathbf{u}_i$ itself, respectively. It is observed that the MAP values of $\lambda_{i,15}$ become relatively small as more data is accumulated, implying that there are no load changes on the 15th floor. Correspondingly, the MAP estimates of $\alpha_{i,15}$ are also relatively small, implying that the load of the 15th floor remains zero. Fig. 5(b) and (d) show that there are large sudden peaks for the MAP values of $\lambda_{i,20}$ and $\alpha_{i,20}$ (the peak values are 4×10^8 and 9.44×10^{11}, respectively) during the time period of 2–2.2 s, which reflects the real situation that the triangular load acts on the 20th floor during

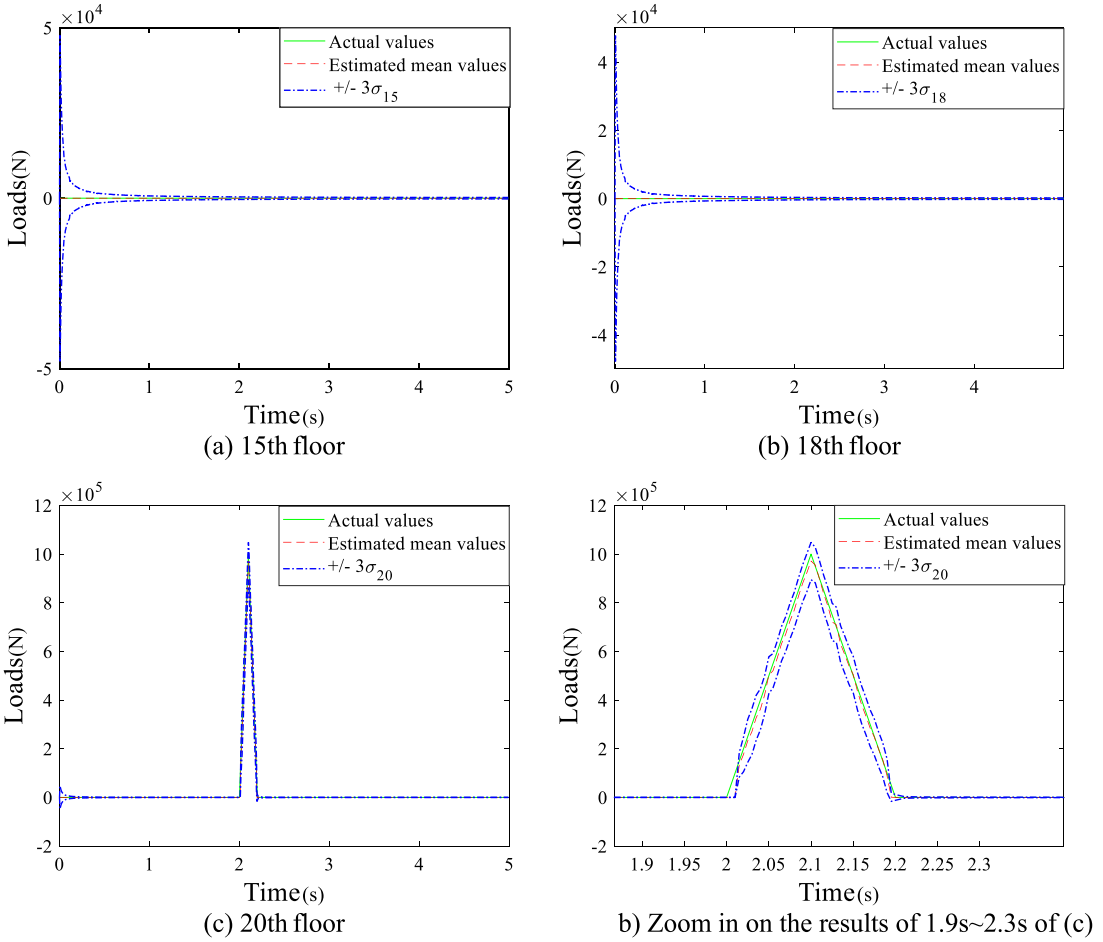


Fig. 2. Comparison of actual and estimated load time histories for the 15th, 18th and 20th floors.

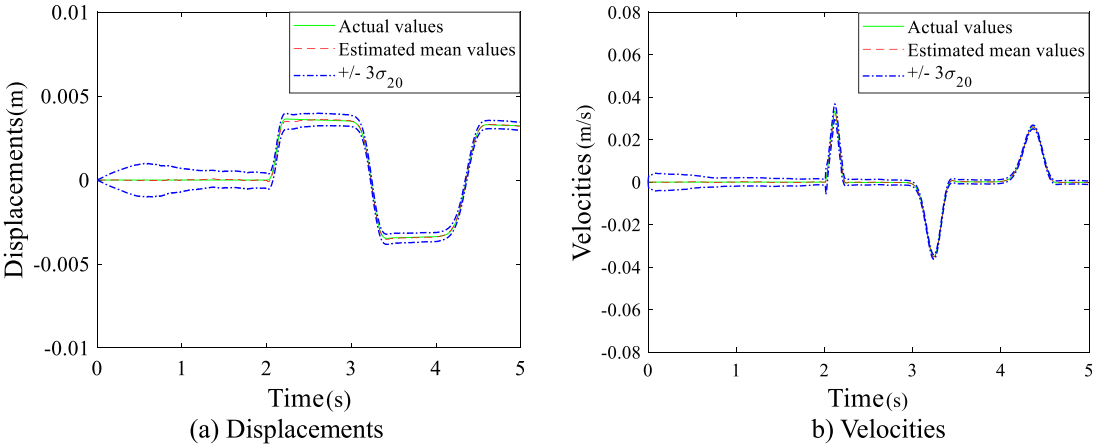


Fig. 3. Comparison of actual and estimated displacement and velocity time histories of the 20th floor.

this time period. The results in Figs. 4 and 5 suggest that the proposed algorithm is able to provide satisfactory estimation of the noise hyper-parameters.

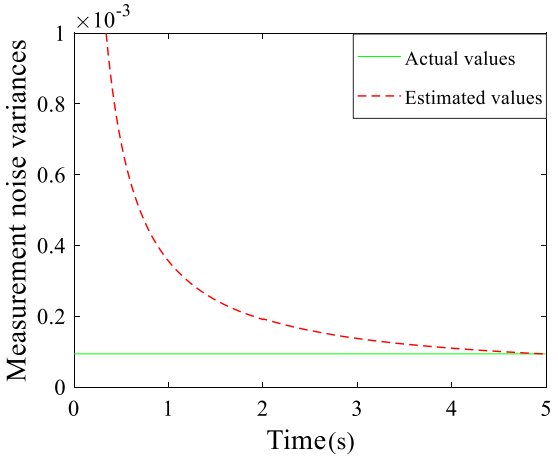


Fig. 4. Comparison of actual and estimated results of the measurement noise variances.

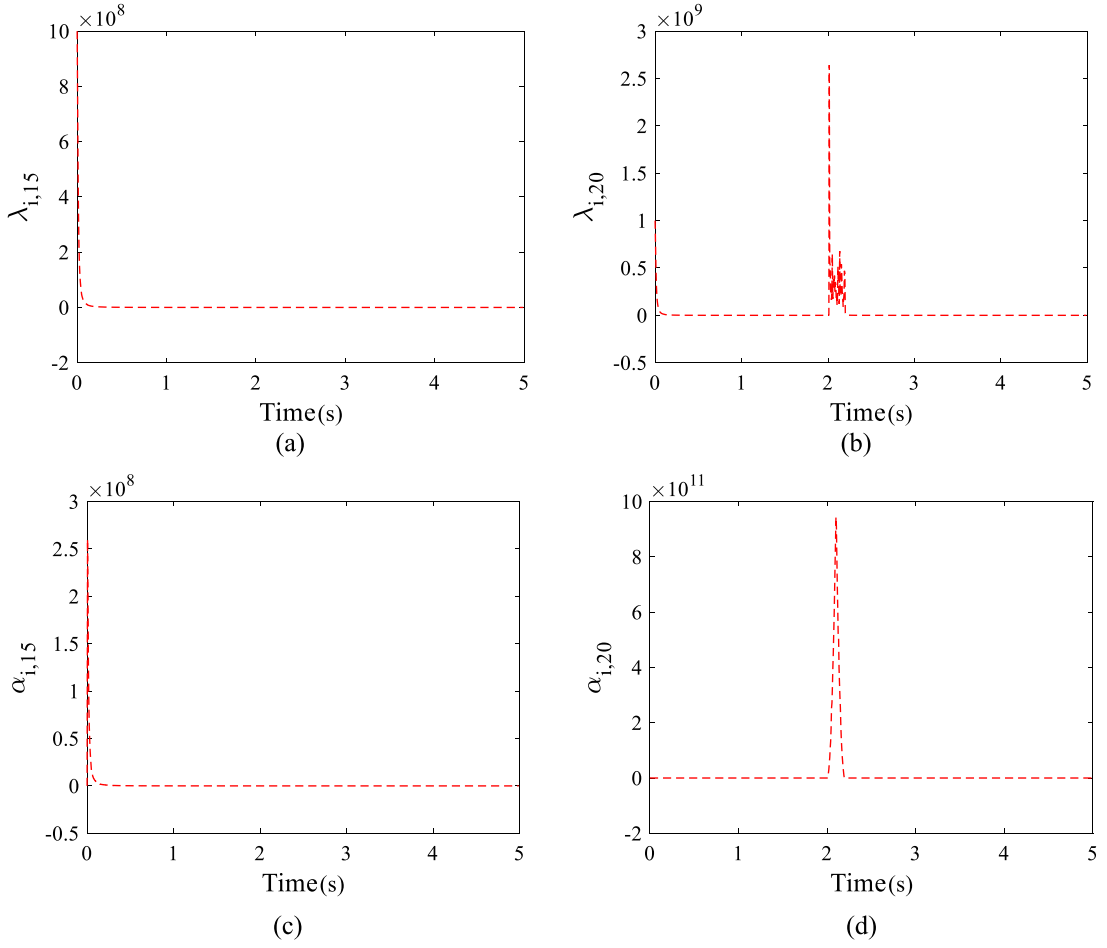
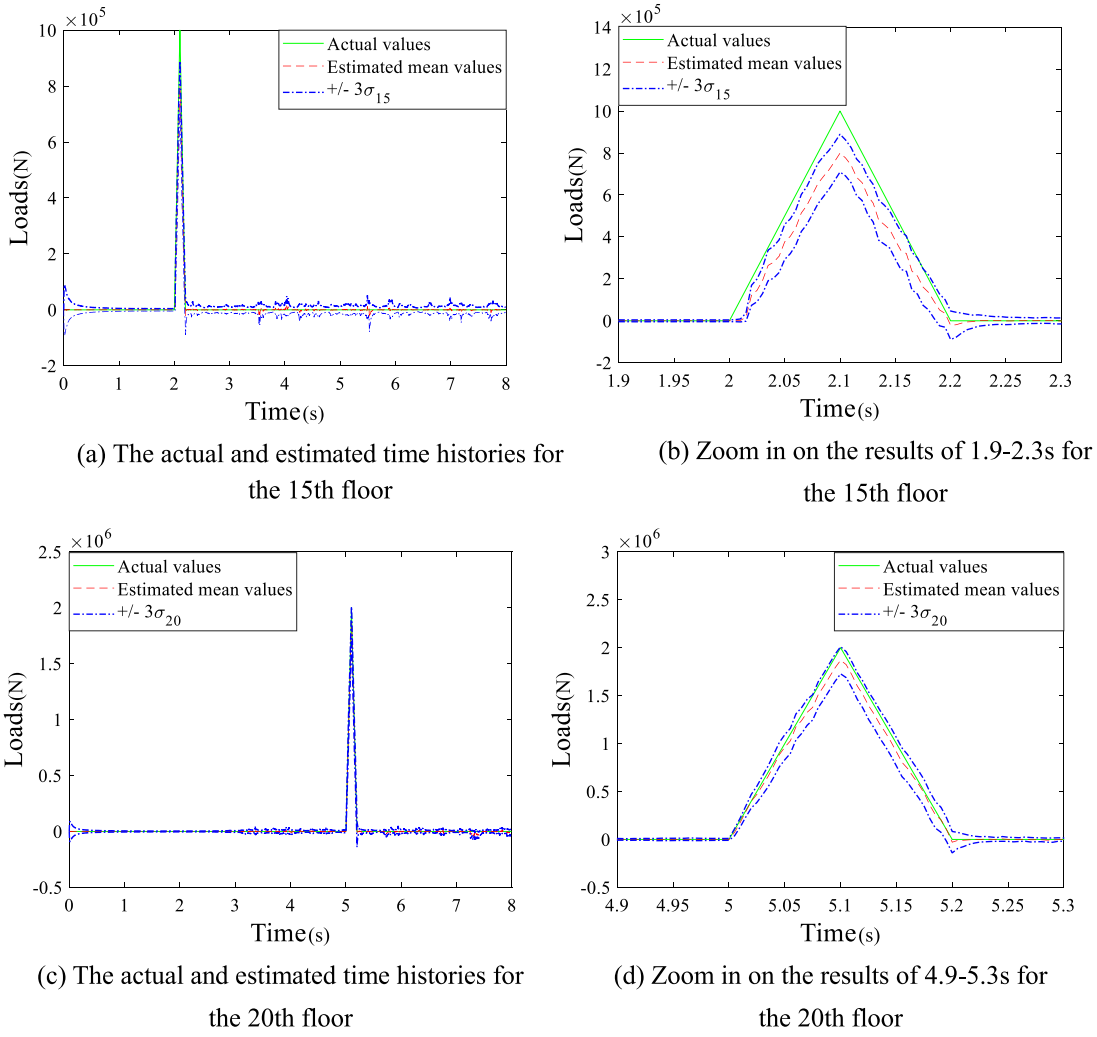


Fig. 5. Comparison of actual and estimated time histories of the MAP values of (*a*) $\lambda_{i,15}$, (*b*) $\lambda_{i,20}$, (*c*) $\alpha_{i,15}$ and (*d*) $\alpha_{i,20}$.



(a) The actual and estimated time histories for the 15th floor

(b) Zoom in on the results of 1.9-2.3s for the 15th floor

(c) The actual and estimated time histories for the 20th floor

(d) Zoom in on the results of 4.9-5.3s for the 20th floor

Fig. 6. Comparison of actual and estimated time histories of impact loads for the 15th and 20th floors.

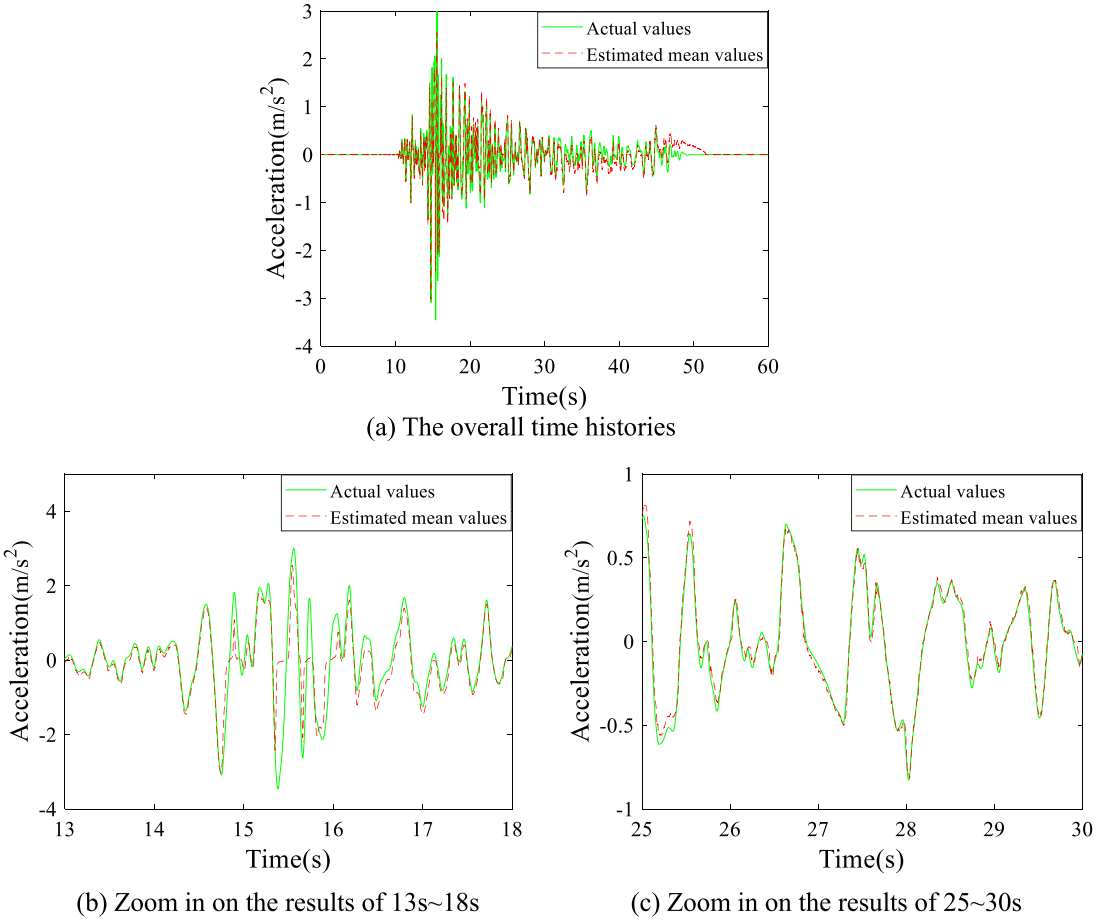
4.2.2. Double-point impact load identification

Double-point impact loads are now considered. The total monitoring time is 8 s and at the time instant of 2 s, the same triangular impact load is applied as before but to the 15th floor, so it reaches to a peak value of 10^3 kN and then falls to zero after 0.1 s and 0.2 s, respectively. Another triangular impact load is applied to the 20th floor over the time period of 5.0–5.2 s with its peak value of 2×10^3 kN at the time instant of 5.1 s.

In Fig. 6, the estimated impact load time histories for the 15th and 20th floors are presented. For both floors, the proposed method can effectively track the changes of impact loads. For the 20th floor, the 99.7% confidence intervals are able to cover the real time histories of the load and the estimated mean of the peak value is 1869.1 kN compared with the real value of 2000 kN. For the 15th floor, the peak of the impact load at 2.1 s lies outside the 99.7% confidence intervals and the estimated mean peak value is 799.5 kN compared with the actual value of 1000 kN, so the impact load estimation is less accurate.

4.2.3. Identification of seismic excitation

In this example, the Humbolt Bay seismic excitation is applied to the structural model for a duration of 40 s, where the peak ground acceleration is 300 gal. In Figs. 7 and 8, the estimated time histories of the seismic excitation and the induced structural states produced by our Algorithm 1 are presented. The estimated time histories agree closely with the actual time histories, demonstrating the excellent performance of the proposed algorithm.



(a) The overall time histories

(b) Zoom in on the results of 13s~18s

(c) Zoom in on the results of 25~30s

Fig. 7. Comparison of actual and estimated time histories of Humbolt Bay seismic excitation.

To show the effect of dropping the soft constraint in Eq. (6) where $\boldsymbol{\theta}_i^c = \mathbf{0}_{N_d}$, the estimation performance of a modified Algorithm 1 is investigated where the variance $\beta^{-1}\alpha_i$ in Eq. (6) is set to a very large value of 10^8. It is interesting to see from Fig. 9 that a drift occurs in the estimated load and displacement time histories in this case, which shows the importance of the soft constraint using the pseudo likelihood function in Eq. (6), where a small value of the hyper-parameter α_i induces the excitation to be close to the reference value of zero.

4.3. Stiffness degradation identification

In this section, the same 20-story shear frame structural model as in Section 4.2 is employed. For stiffness identification purposes, the uncertain structural stiffness scaling parameter $\mathbf{w}_i \in \mathbb{R}^{N_w}$ will be the model parameter vector $\boldsymbol{\theta}_i$ that is tracked, together with the state vector, by using Algorithm 1. The same Humbolt Bay seismic excitations as in Section 4.2.3 is applied to the structural model. A 10% stiffness loss occurs for $\boldsymbol{\theta}_i$ for the 1st story over the time period [15,15.1] s. Instrumentation corresponding to both complete and incomplete sets of observations are investigated, where for the incomplete observation set, ten accelerometers are placed at the locations corresponding to the 3rd, 5th, 7th, 8th, 10th, 13th, 15th, 17th, 18th and 20th floors, i.e. , $N_m = 10$. For the complete observation set, an accelerometer is placed on each of the 20 floors. As in Section 4.2, the sensor sampling frequency is 200 Hz and Gaussian measurement noise with standard deviation equal to 10% of RMS of the roof acceleration is added to the acceleration data.

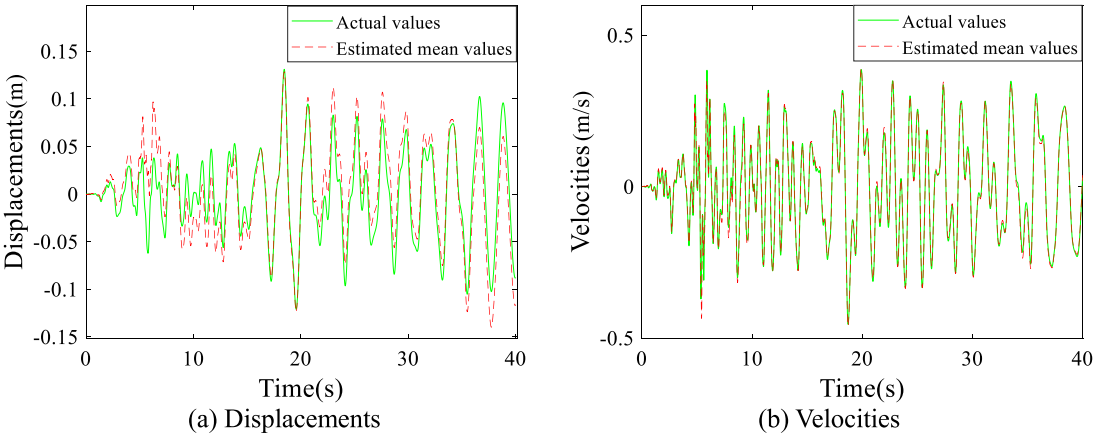


Fig. 8. Comparison of actual and estimated displacement and velocity time histories at the 20th floor.

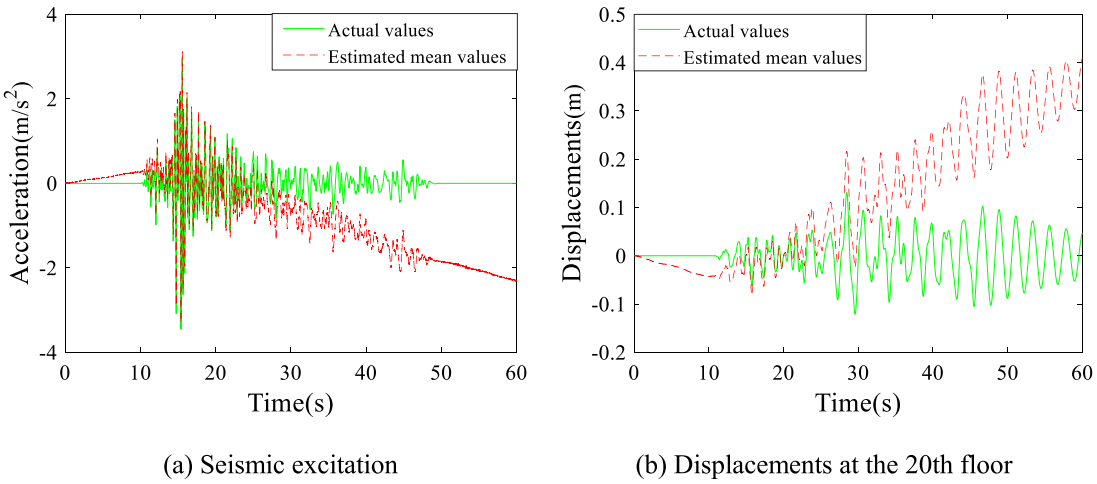


Fig. 9. Comparison of actual and estimated seismic excitation and structural displacement time histories when fixing the variance $\beta^{-1}\alpha_i$ in Eq. (6) to the value of 10^8.

4.3.1. Stiffness degradation identification under known loads

For this example, the loads on the structure are assumed known and the estimated stiffness degradation results for the 1st, 3rd and 20th story are presented in Fig. 10. We first compare the results based on the complete and incomplete sets of observations in Fig. 10(i) and (ii), respectively. For the incomplete observation set, it is observed that there is a longer time delay before the stiffness degradation is detected and the posterior uncertainties are slightly larger than for the complete observation set; however, the actual stiffness evolution is still within the 99.7% confidence intervals (± 3 posterior standard deviation) for most of the monitoring periods except for 5 s to 10 s immediately after the abrupt change in $\boldsymbol{\theta}_i$. In order to validate the usefulness of the sparseness constraint for $\Delta\boldsymbol{\theta}_i = \boldsymbol{\theta}_i - \boldsymbol{\theta}_i^c$ produced by the introduction of the pseudo-likelihood function in Eq. (6), we also show results in Fig. 10(iii) where the variances $\beta^{-1}\alpha_{i,n}$ in Eq. (6) are fixed to be 10^8, i.e., we weaken the constraint that $\Delta\boldsymbol{\theta}_i$ should be a sparse vector. It is seen that the estimated mean values of the stiffness parameters become less stable and the 99.7% confidence intervals are obviously larger, indicating larger uncertainty in the estimation and lower accuracy of the posterior mean. For the 20th story, the estimated stiffness drifts away from the actual values with increasing time. Fig. 11 corresponds to the estimated displacement and velocity time–history curves at the 20th floor. The results demonstrate the effectiveness of the state vector estimation.

To show the performance if the stiffness loss occurs in an upper floor, the same configuration as that in Fig. 10 is employed but 10% stiffness loss is assumed for the 6th story over the time period [15,15.1] s and the results are

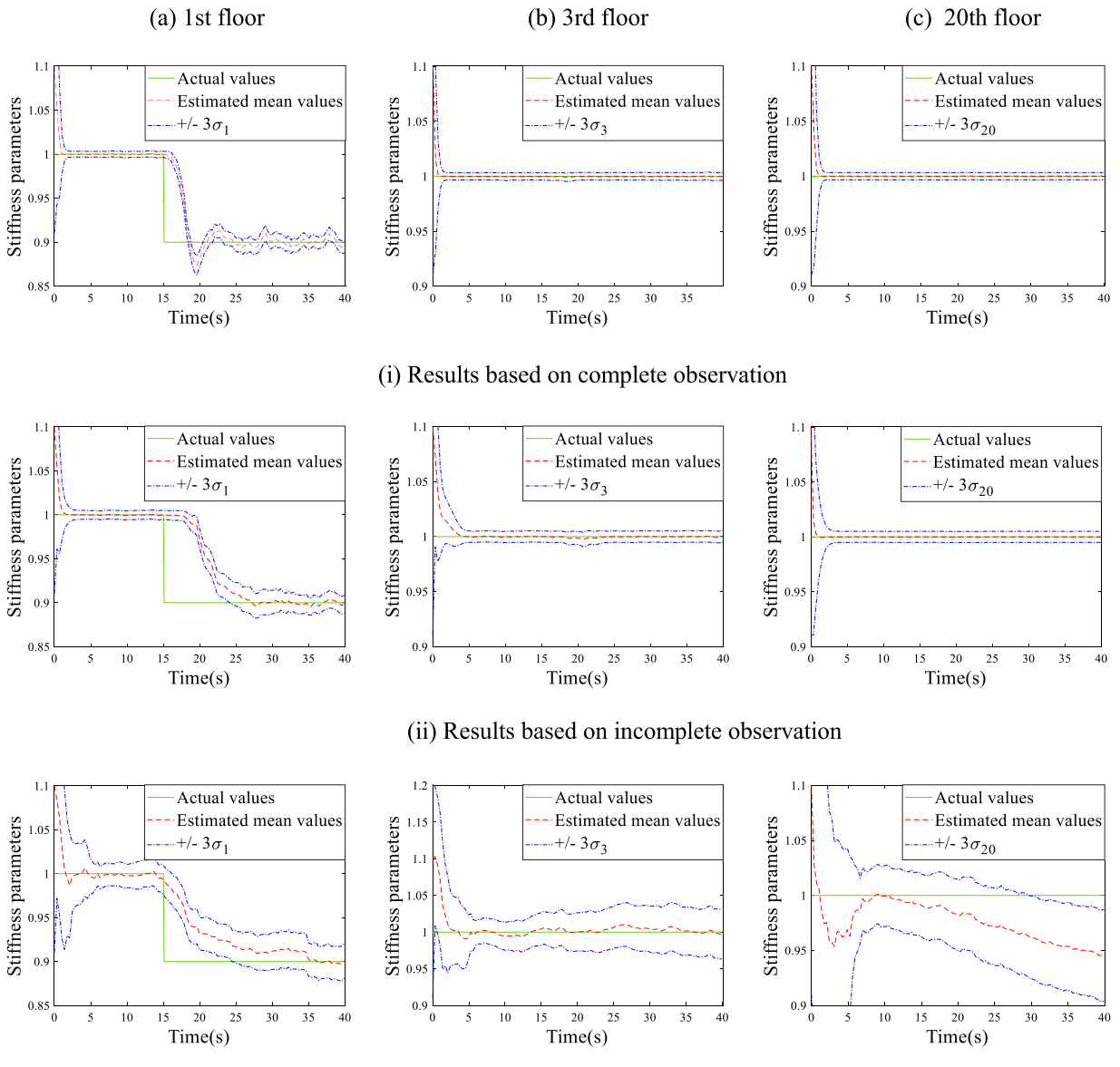


Fig. 10. Estimated time histories of stiffness parameters for (a) 1st, (b) 3rd and (c) 20th floors under the Humbolt Bay seismic excitation when a 10% stiffness loss occurs for the 1st story over the time period [15,15.1] s.

reported in Fig. 12. It is observed that the actual stiffness evolution is within the 99.7% confidence intervals (±3 posterior standard deviation) for most of the monitoring periods, though there are slighter longer time delays (5–10 s and 10–15 s for complete and incomplete observations, respectively) before the stiffness degradation is detected compared with those of Fig. 10. The algorithm demonstrates excellent performance no matter whether the stiffness loss occurs on the lower or upper floors.

4.3.2. Stiffness degradation identification under unknown loads

For a real structural system, both the dynamic loads and stiffness degradation would usually be unknown and so they are required to be identified simultaneously together with the evolution of the structural states. The same

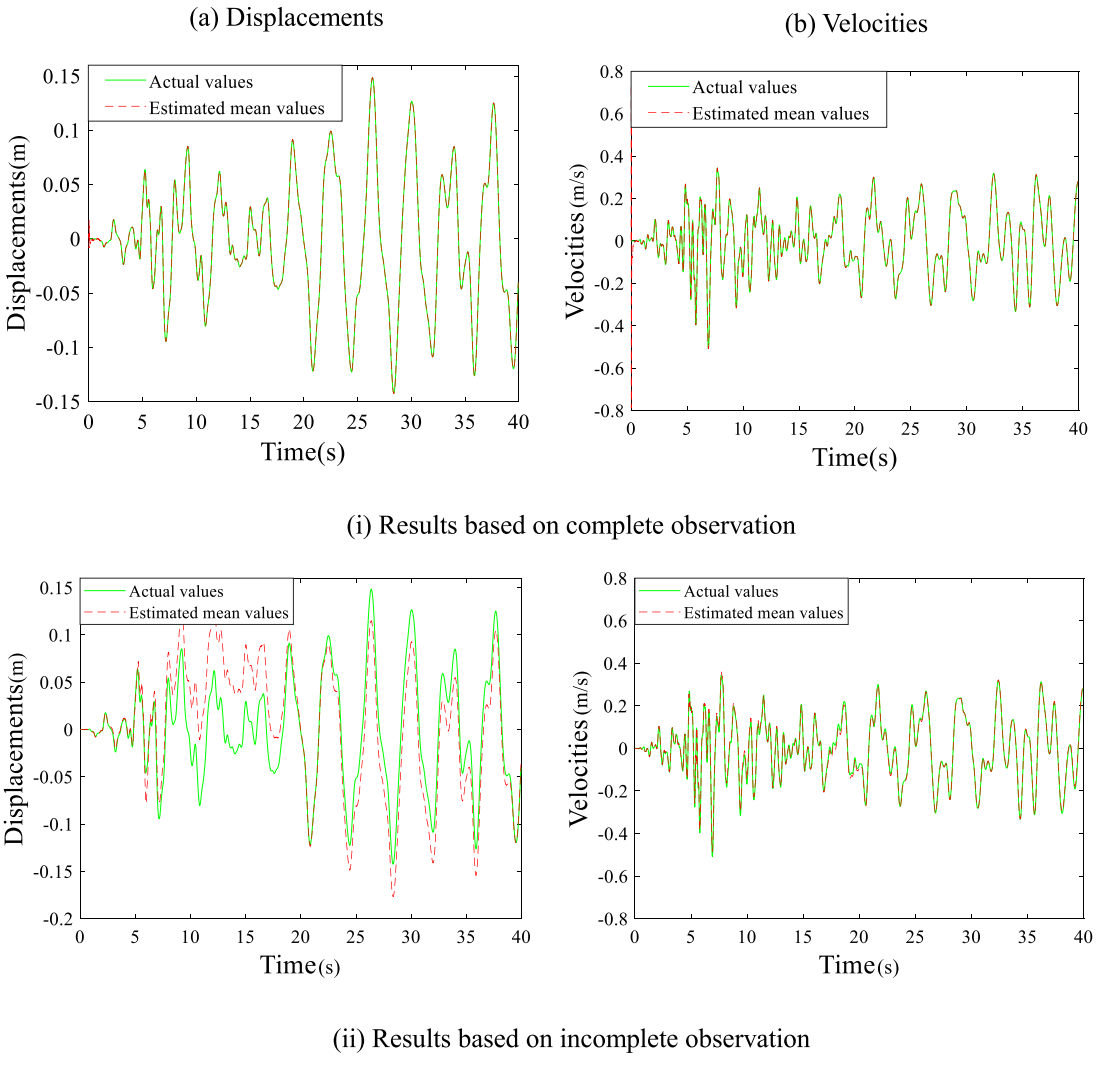


Fig. 11. Estimated (a) displacement and (b) velocity time histories of the 20th floor under random excitation based on noisy observation when a 10% stiffness loss occurs for the 1st story over the time period [15,15.1] s.

configuration as that in Section 4.3.1 is employed but under unknown seismic excitation. The identification results for the dynamic loads and stiffness degradation are presented in Figs. 13 and 14, respectively.

It is seen that in this case the seismic excitation time histories can still be estimated accurately, as the actual and estimated curves almost exactly coincide. In addition, the stiffness degradation at the 1st floor can be estimated effectively, although compared with the results of the stiffness estimation under known loads in Fig. 10, the tracking of the stiffness degradation is not as accurate. This is due to the extra ill-posedness induced by the simultaneous estimation of the loads and stiffnesses. Fig. 15 corresponds to the estimated displacement and velocity time histories, which show that the displacements and velocities are estimated effectively in this case.

4.4. Further investigation of the algorithm performance

In this section, the performance of our Algorithm 1 is further verified using a more complex damping model, various numbers of sensors and various noise levels. The same 20-story shear frame structural model as in Sections 4.2 and 4.3 is employed. The Humbolt Bay seismic excitation is again applied to the structural model for a duration of 40 s, where the peak ground acceleration is 300 gal.

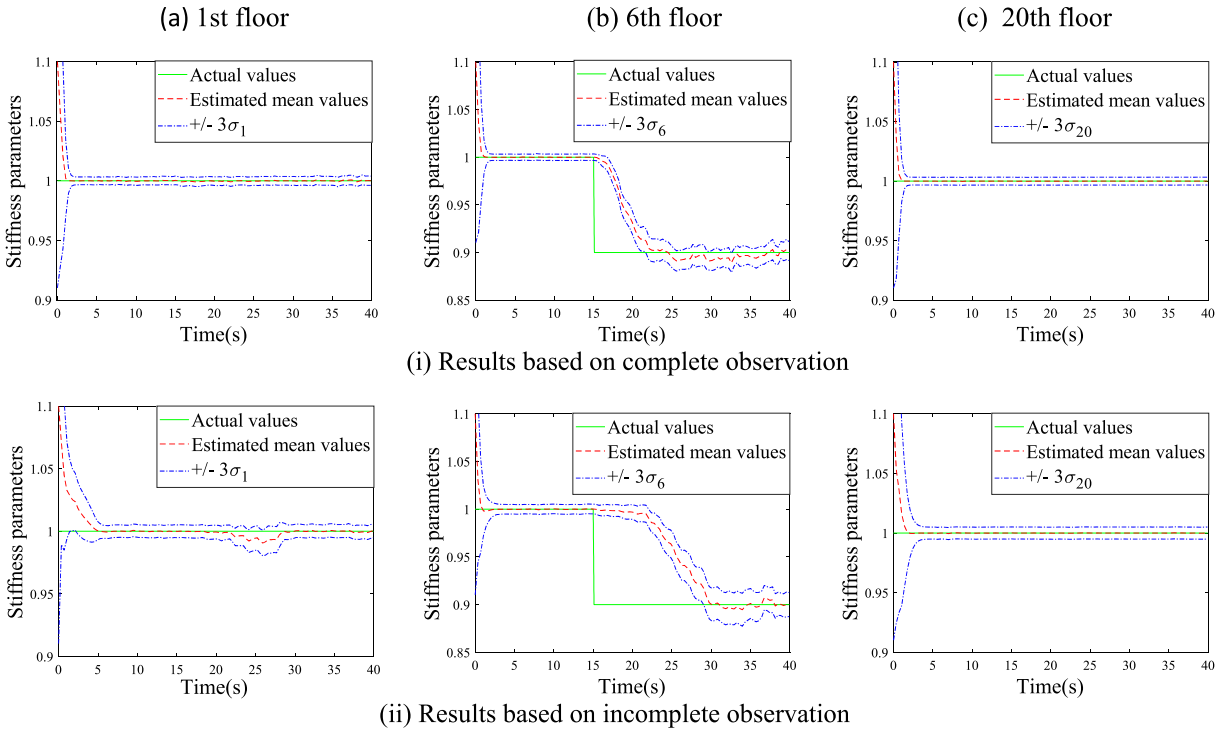


Fig. 12. Estimated time history curves for stiffness parameters of (a) 1st, (b) 6th and (c) 20th floors for the Humbolt Bay seismic excitation when a 10% stiffness loss occurs for the 6th story over the time period [15,15.1] s.

4.4.1. Algorithm performance with a more complex damping model

In this section, Caughey series damping, $\mathbf{C}_s = \mathbf{M}_s \sum_{l=0}^{L-1} a_l \left[\mathbf{M}_s{}^{-1}\mathbf{K}_s\right]^l$, is adopted to represent a generalization of the Rayleigh damping matrix and we select the number of terms in the series as $L = 4$. The first four damping ratios associated with the Caughey damping matrix are chosen as 2%, so that the coefficients $a_l{}'s$ $(l = 0, \ldots, 3)$ are $a_0 = 0.0764$, $a_1 = 0.0047$, $a_2 = -1.53 \times 10^{-5}$ and $a_3 = 2.093 \times 10^{-8}$, respectively. In Fig. 16, the estimated time histories of the seismic excitation based on the incomplete observation set are presented, where the same configuration as that in Section 4.3.1 is employed but using the Caughey damping model. It is seen that overall the estimates effectively track the actual time history, although compared with the results in Fig. 7 where a Rayleigh damping matrix (corresponding to $L = 2$ in $\mathbf{C}_s$) is used, the tracking of the seismic excitation is not as accurate and deviations from the actual values occur at some peak values, which is presumably because of the higher complexity of the damping model.

We note that our algorithm cannot be used for tracking stiffness changes when the Caughey damping matrix with $L = 4$ is employed, since we cannot express the measurement $\mathbf{y}(t)$ linearly with respect to the stiffness scaling parameter vector $\mathbf{w}$ directly as in Eq. (32), even though the structural stiffness matrix $\mathbf{K}_s$ is modeled as a linear function in terms of $\mathbf{w}$. The possible solution of this difficulty is direct linearization of the measurement $\mathbf{y}(t)$ with respect to $\mathbf{w}$ for each time step, which will be of our future work.

4.4.2. Algorithm performance using various numbers of accelerometers

In system identification, if only a small number of accelerometers is available, it may occur that the sensor data $\mathbf{y}_i$ provides insufficient information to accurately track time-varying parameter values. To study this effect, the results of separate seismic excitation identification and stiffness change identification are presented in Figs. 17 and 18, respectively, using measurements from 5 accelerometers (one on each of the 3rd, 6th, 10th, 15th, and 20th floors, i.e. $N_m = 5$) and 7 accelerometers (one on each of the 3rd, 6th, 10th, 13th, 16th, 18th and 20th floors, i.e. $N_m = 7$). It is seen that the seismic excitation can be tracked effectively when 7 accelerometers are available. However, when the number of accelerometers is only 5, the time histories of seismic excitation are significantly underestimated,

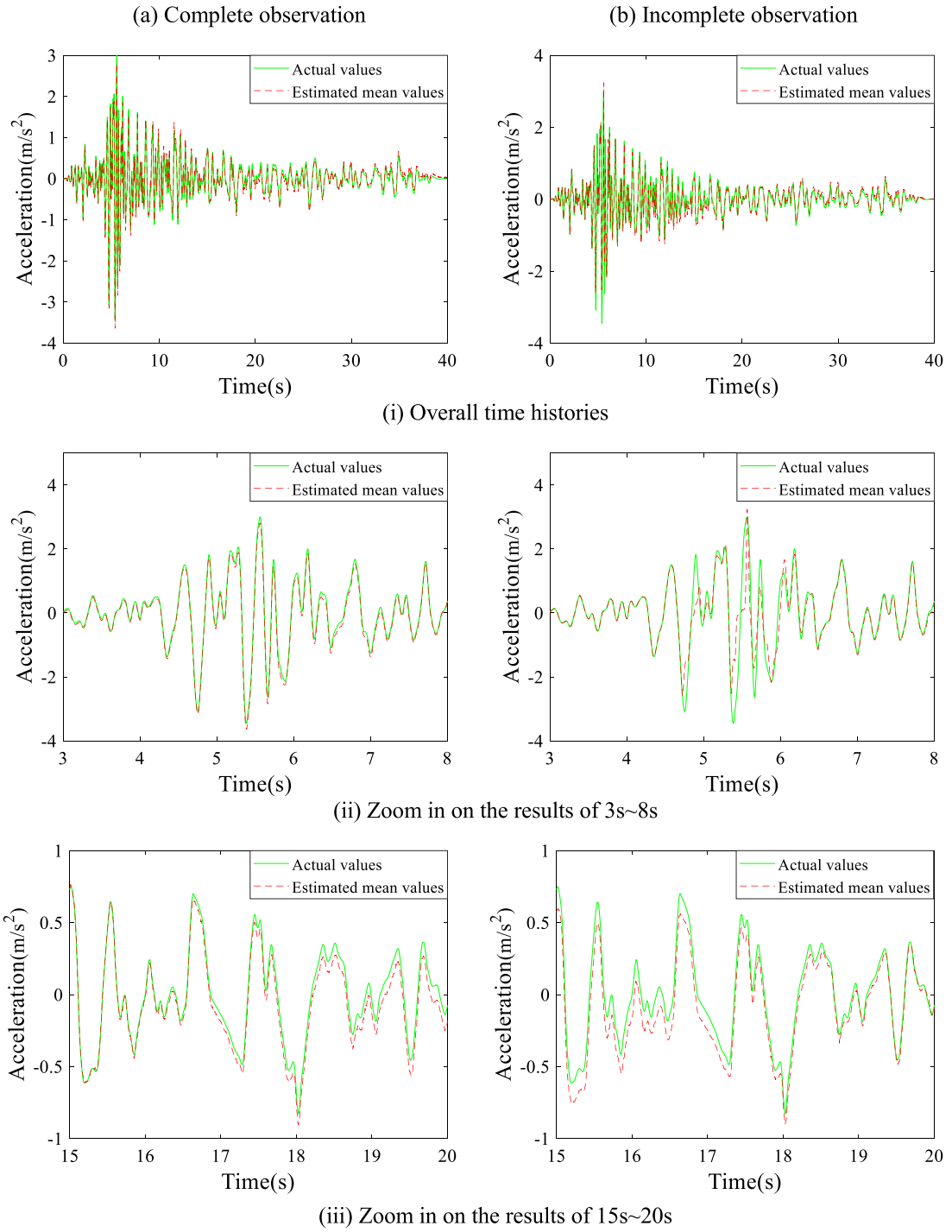


Fig. 13. Time history curves for the Humbolt Bay seismic excitation.

demonstrating poor performance of the proposed algorithm for input identification when using only this amount of data.

For stiffness identification, however, both the 5 and 7 accelerometer data sets produce good estimates since the actual stiffness evolution is still within the 99.7% confidence intervals (± 3 posterior standard deviation) for most

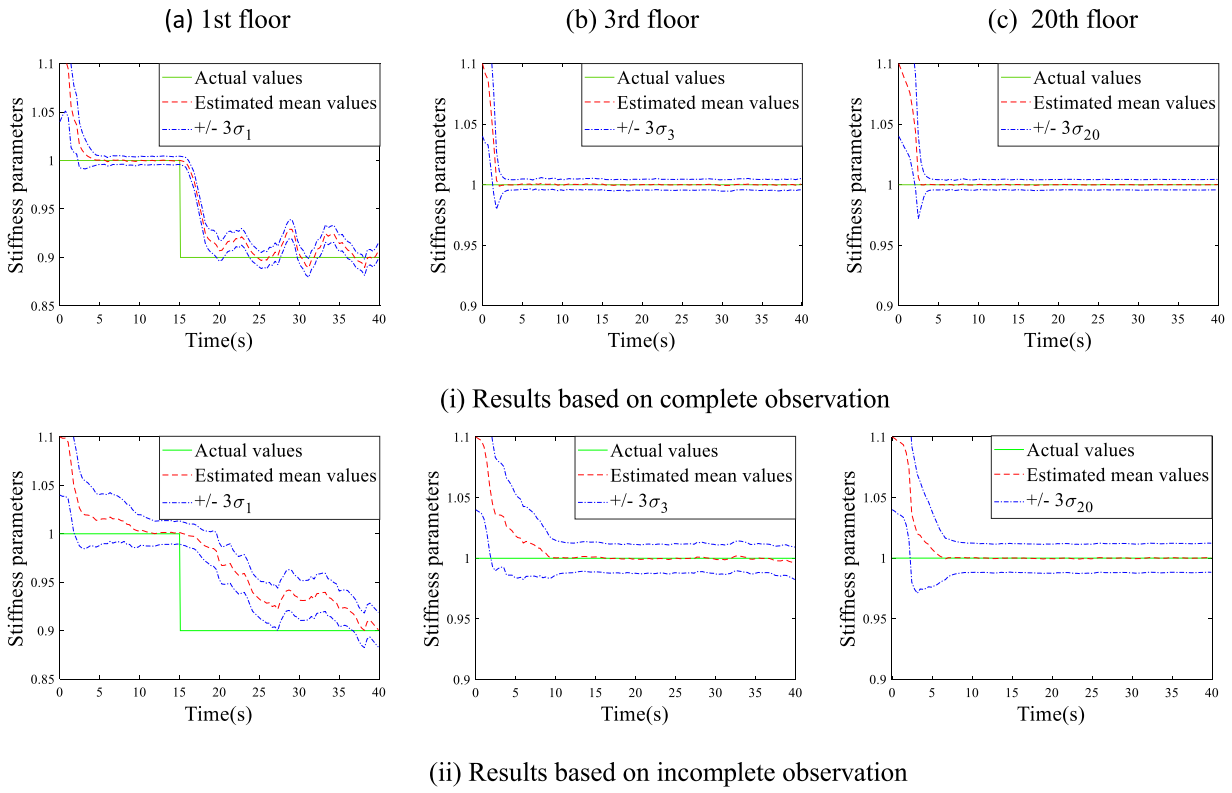


Fig. 14. Estimated time history curves for stiffness parameters of (a) 1st, (b) 3rd and (c) 20th floors under the assumed unknown Humbolt Bay seismic excitation.

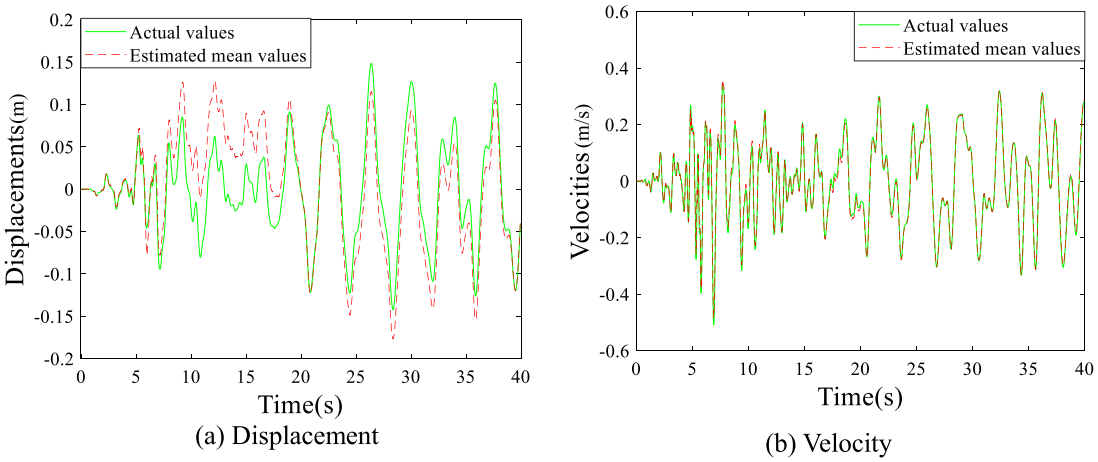
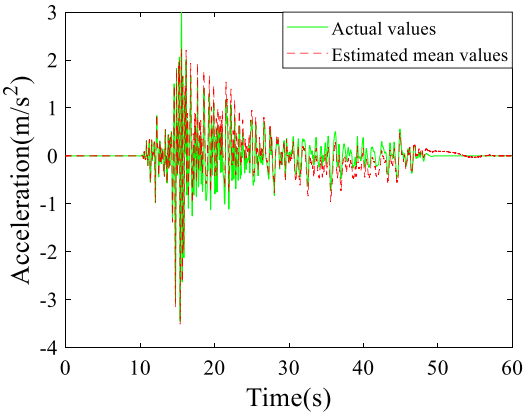


Fig. 15. Displacement and velocity time histories of the 20th floor under the assumed unknown Humbolt Bay seismic excitation.

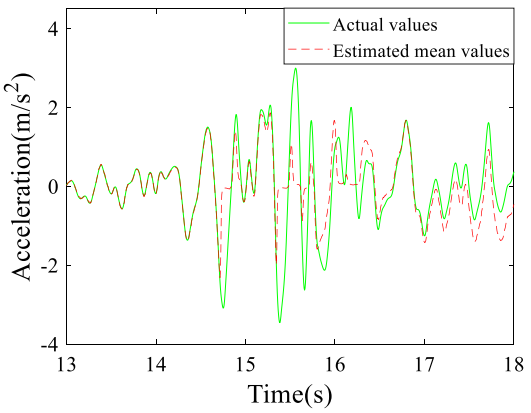
of the monitoring periods, even though there are longer time delays (see Fig. 18) before the stiffness degradation is detected compared with the results with 10 accelerometers available (see Fig. 10). Therefore, it can be concluded that the method performs effectively even when only 5 accelerometers are available.

4.4.3. Algorithm performance under various noise levels

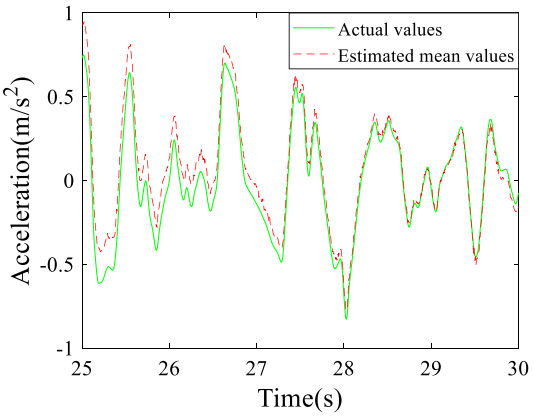
Zero-mean Gaussian white noise with standard deviations equal to 5% and 15% of the RMS of the roof acceleration, instead of 10% used previously, were added to the previous incomplete set of acceleration responses.



(a) The overall time histories



(b) Zoom in on the results of 13s~18s



(c) Zoom in on the results of 25~30s

Fig. 16. Comparison of actual and estimated time histories of Humbolt Bay seismic excitation using a Caughey damping model (the number of terms in the series is $L = 4$).

Then identification was performed separately for the time histories of seismic excitation and stiffness evolution and the results are presented in Figs. 19 and 20, respectively. The same configuration as that under 10% noise in Fig. 7 is employed for the different noise levels. Fig. 19 shows that for 5% noise, the estimated seismic excitation time histories agree closely with the actual time histories while for the 15% noise, the time histories deviate significantly from the actual histories for some time instants, which is consistent with our expectation.

By observing the stiffness identification results under 5% and 15% noise levels in Fig. 20 and that under 10% noise in Fig. 10, it is seen that our algorithm produces accurate stiffness tracking results for all three noise levels. However, there is a longer time delay occurring before the stiffness degradation is detected as the noise level increases.

4.5. Real data example

In this subsection, the performance of sequential state and input estimations of the proposed algorithm is verified by using shaking table data. The tests were conducted on a 5-story small-scale shear structure that consists of two vertical aluminum plates and six thick plexiglass plates, as shown in Fig. 21 [47]. A low-pass white-noise excitation with a cut-off of 200 rad/s is employed to shake the test structure, and the structural response with a sampling rate of 200 Hz is collected by five accelerometers, one on each floor. The monitoring time is 40 s. Before the state and input estimations, the structural story stiffness and damping can be identified from the measured structural

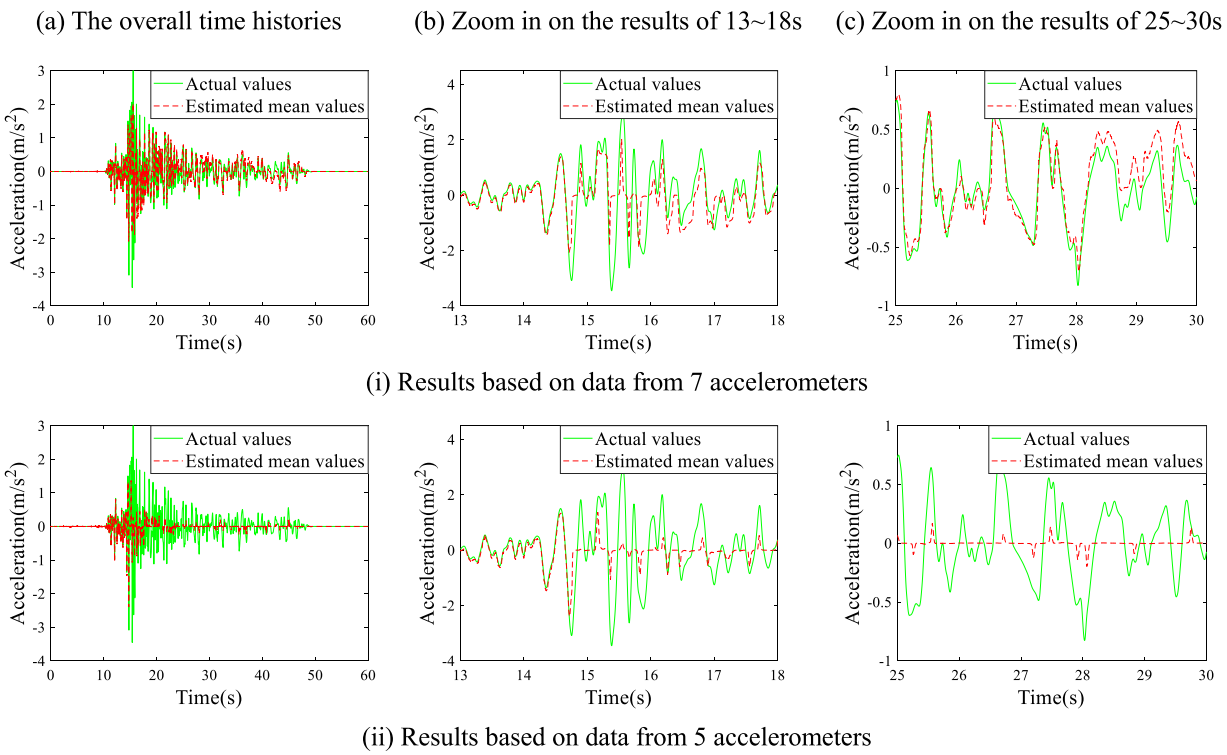


Fig. 17. Humbolt Bay seismic excitation identification results based on the measured accelerations from different numbers of sensors.

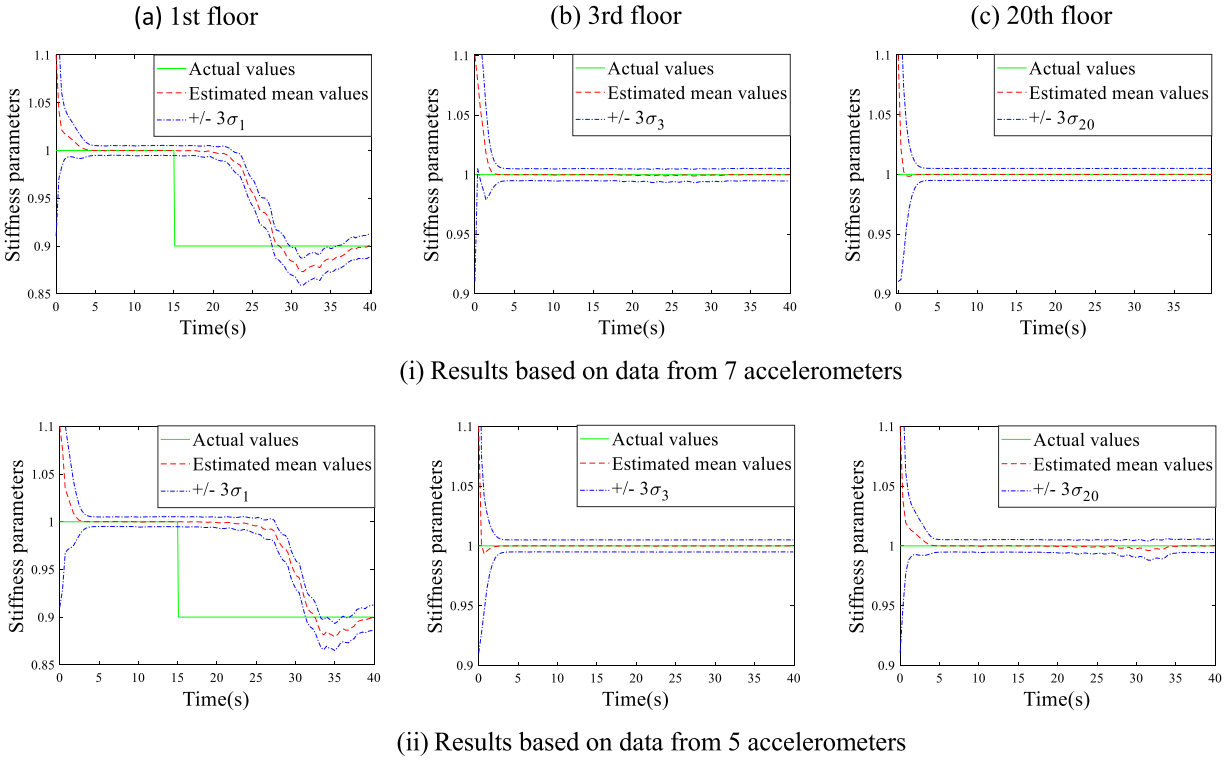


Fig. 18. Estimated time history curves for stiffness parameters of (a) 1st, (b) 3rd and (c) 20th floors under the Humbolt Bay seismic excitation based on the response from different numbers of accelerometers.

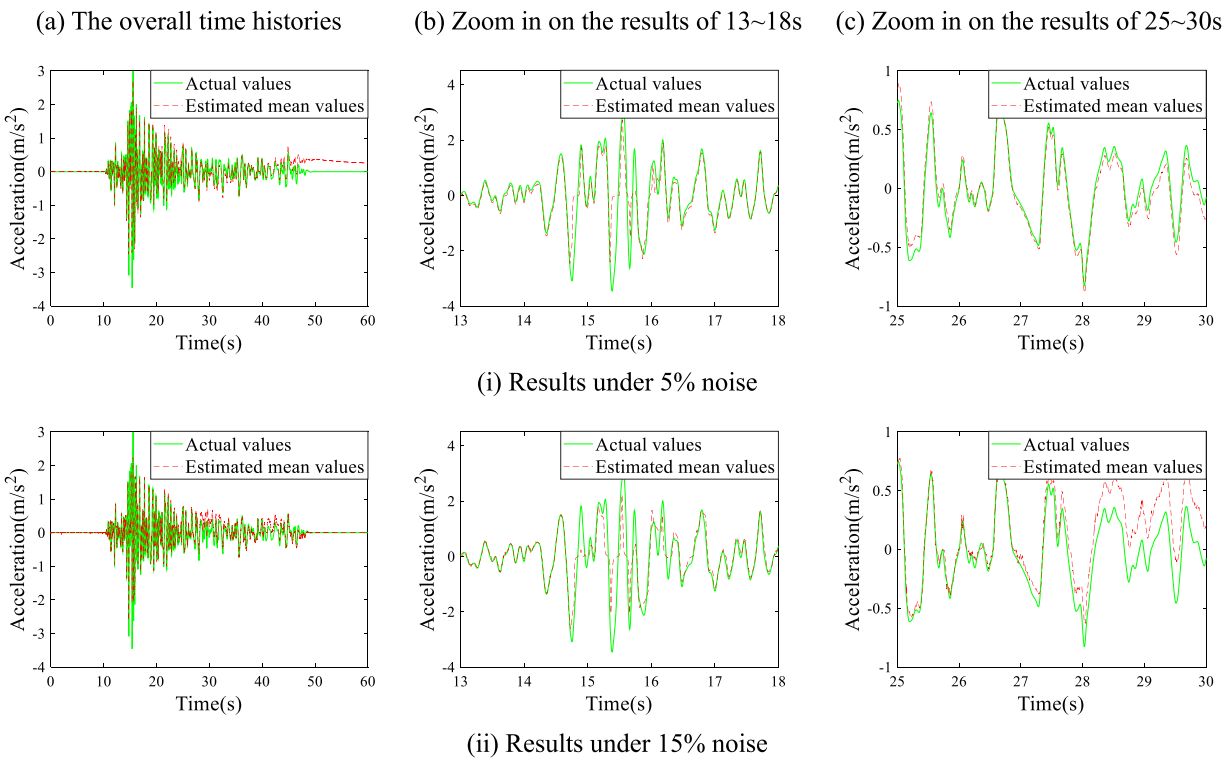


Fig. 19. Estimated time histories of Humbolt Bay seismic excitation results under various noise levels.

responses, which can then be used as the structural model parameter values together with the nominal structural mass during the sequential state and input identification. For detailed information about the structural model and experimental configuration, as well as the model parameter estimation, the reader can refer to [47].

The estimated time histories of the structural excitation and the associated structural states from applying the proposed algorithm are presented in Figs. 22 and 23. It is obvious that the actual excitation can still be tracked accurately by our algorithm in the case of complete observation using accelerometer data from all floors, as the actual and estimated curves coincide almost exactly. For incomplete observation using three accelerometers, one on each of the 2nd, 4th and 5th floors, the excitation can also be estimated effectively, although compared with the previous results for the numerical examples, the tracking of the excitation is not as accurate. This inaccuracy is due to the extra modeling error between the real structural behavior and that of the chosen underlying structural model. Overall, our algorithm performs well in this experimental example.

5. Conclusion

In this paper, simultaneous tracking of the system states and model parameters for dynamic systems is investigated. A dual Kalman filter method is proposed for real-time Bayesian sequential state and parameter estimation based on noisy acceleration signals. In this method, soft (probabilistic) sparseness constraints for changes of model parameter from their initial reference values as well as changes of the model parameter between two successive time instants are imposed by a hierarchical sparse Bayesian learning model. Analytical Kalman filtering equations can be derived because our stochastic model of the structural dynamical system can be represented as a coupled conditionally-linear Gaussian state space model for the state and model parameter vectors. In addition, the parameters for the measurement and state prediction errors are learned solely from the available data up to the current time, which is useful to resolve the well-known instability problem in Kalman filtering due to arbitrary assignment of the error-distribution parameters.

Two illustrative applications are presented: one for load time–history estimation and the other for identification of stiffness degradation where both are based on noisy incomplete dynamic response measurements from a structure.

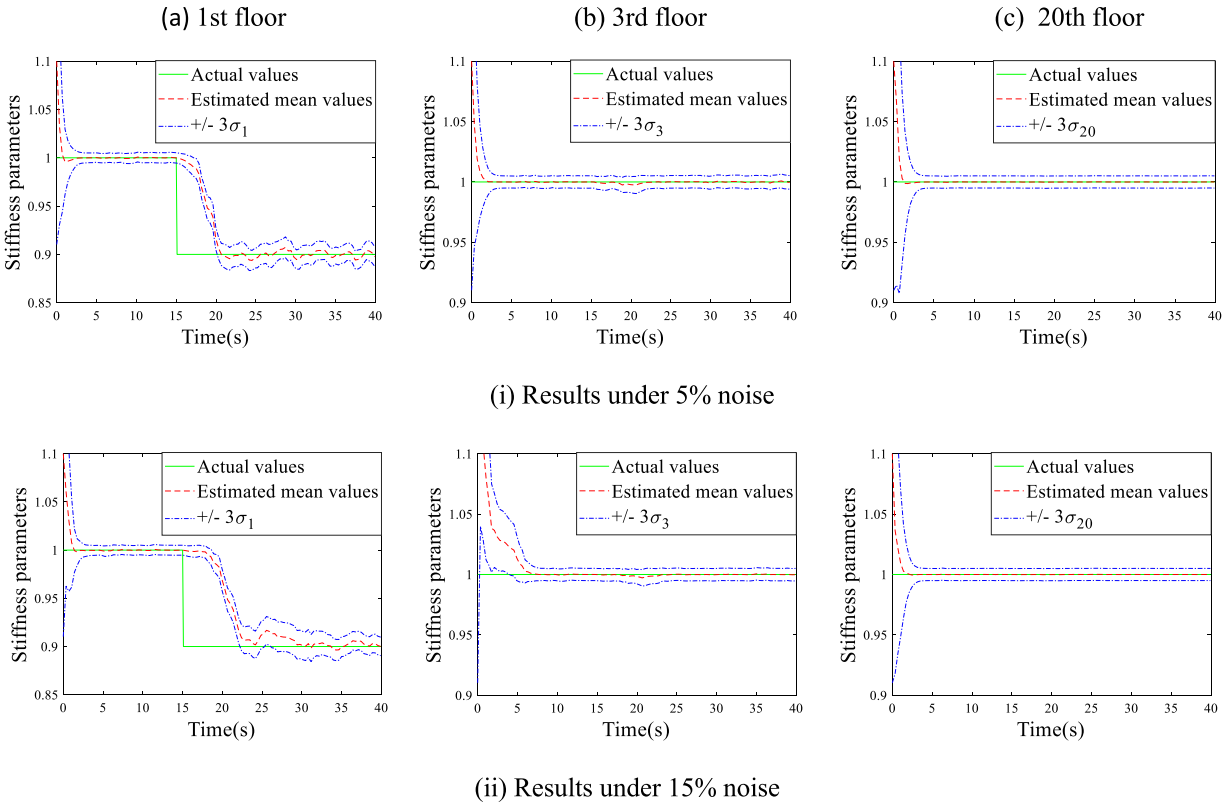


Fig. 20. Estimated time histories of the stiffness parameters for (a) 1st, (b) 3rd and (c) 20th floors under the Humbolt Bay seismic excitation under various noise levels.



Fig. 21. Test structure in shake table tests.

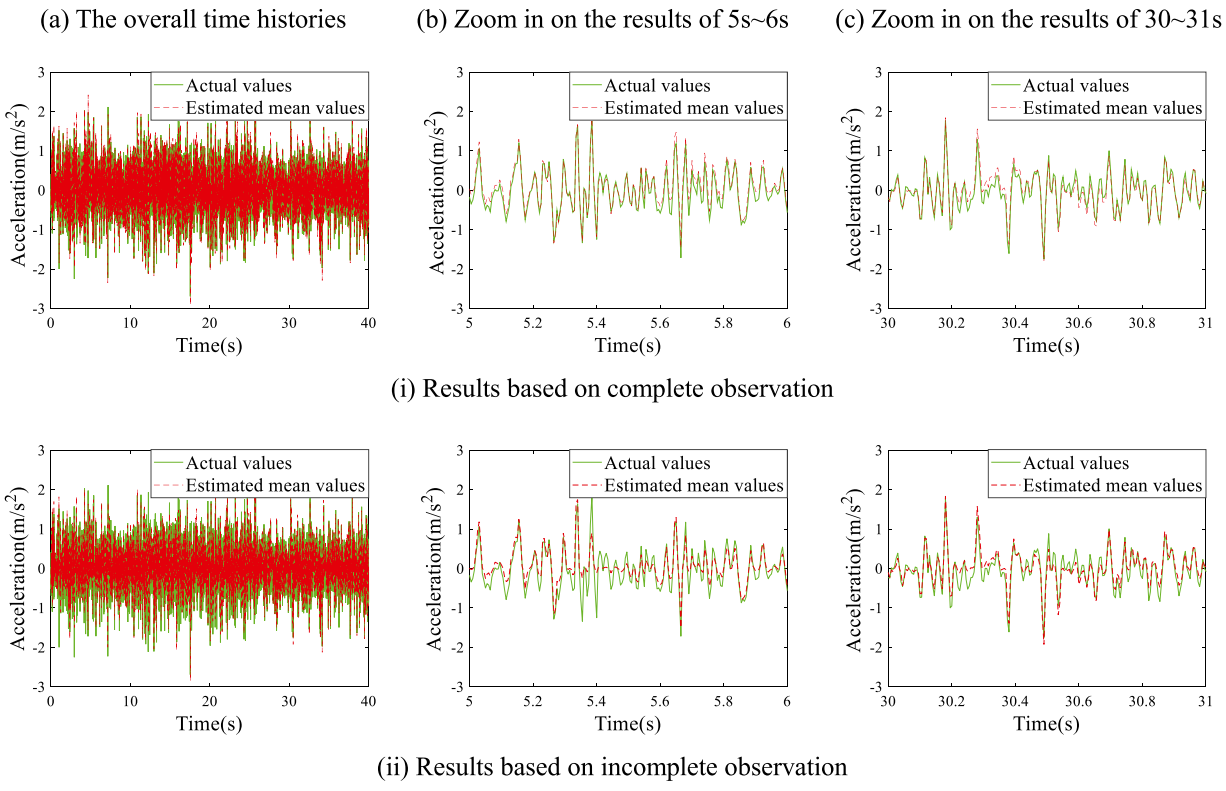


Fig. 22. Comparison of actual and estimated time histories of excitation.

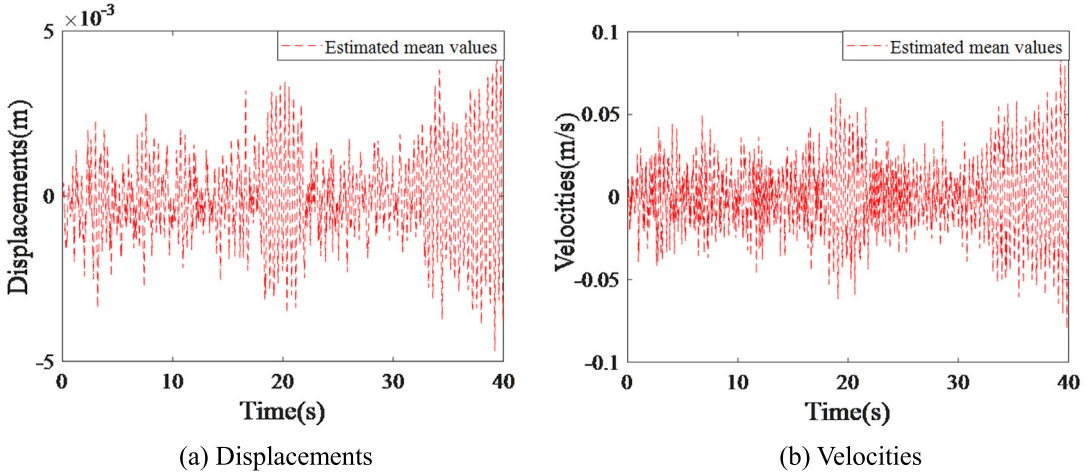


Fig. 23. Estimated displacement and velocity time histories at the 5th floor based on incomplete observation.

The proposed dual Kalman filter algorithm performed satisfactorily for all examples, judging from the fact that it produces estimates for the system state and unknown load or stiffness parameters with associated confidence intervals that are usually consistent with their actual values. Comparing the two applications, the estimation of external loads is relatively more accurate because there is a time delay in the stiffness degradation estimation immediately after an abrupt change of stiffness occurs. The usefulness of the pseudo-likelihood function which imposes the soft constraint that model parameter changes from their initial reference values should be sparse (for

a multi-dimensional model parameter, such as the stiffness parameters) or should be small (for a single dimension model parameter, such as the seismic load parameter) is also investigated. It is found that the commonly encountered drift phenomenon in load estimation can be effectively suppressed and the accuracy and stability of the stiffness identification can be increased by the proper learning of the prediction error hyper-parameter in the pseudo-likelihood function. The effectiveness of the proposed method for online tracking of the dynamic sparseness of both stiffness losses and structural input is demonstrated. Finally, the algorithm is applied to experimental data from a shake table test conducted on a five-story shear structure for both state and input estimation. It is found that the actual excitation can still be tracked effectively for both complete and incomplete observation data sets, as expected, although the estimation of the excitation is not as accurate for the incomplete observation data set.

In future studies, it would be useful to explore improvements of the algorithm performance for simultaneous input and stiffness identification, together with tracking the evolution of the structural states, which requires development of effective strategies to alleviate the ill-posedness induced by the simultaneous estimation of the loads and stiffnesses. In addition, the algorithm will be extended to utilize a particle filter for Bayesian state and parameter estimations so that it is applicable to highly nonlinear structural models with non-Gaussian uncertainties.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research was supported by grants from the National Natural Science Foundation of China (NSFC Grant Nos. 51778192 and 51638007).

Appendix A. Some implementation details of dual Kalman filter modeling

For state and parameter estimation, the correlation between the state vector and the model parameters must be accounted for. Special care is taken when deriving the observation matrices of the state space model and the predictive PDF and likelihood functions for $\mathbf{x}_i$ and $\boldsymbol{\theta}_i$.

(1) The observation matrices

For the dual EKF equations, there is actually a necessary modification of the linearization $\mathbf{B}_i = \frac{\partial h_i(\boldsymbol{\mu}_{\mathbf{x}_i}^+, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}}$ associated with the filter. This is due to the fact that the filter, whose parameters are being estimated, has a recurrent architecture, i.e., $h_i(\boldsymbol{\mu}_{\mathbf{x}_i}^+, \boldsymbol{\theta})$ is a function of $\boldsymbol{\mu}_{\mathbf{x}_i}^+$, and also a function of $\boldsymbol{\theta}$. Thus, the linearization must be computed using recurrent derivatives with a routine. Similarly, there is also a necessary modification for $\mathbf{C}_i = \frac{\partial h_i(\mathbf{x}, \boldsymbol{\mu}_{\boldsymbol{\theta}_i}^+)}{\partial \mathbf{x}}$. Then the observation matrices $\mathbf{B}_i$ and $\mathbf{C}_i$ of the state space model in (2) are modified as:

$$\mathbf{B}_i' = \frac{\partial h_i(\boldsymbol{\mu}_{\mathbf{x}_i}^+, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}} + \frac{\partial h_i(\boldsymbol{\mu}_{\mathbf{x}_i}^+, \boldsymbol{\theta})}{\partial \boldsymbol{\mu}_{\mathbf{x}_i}^+} \cdot \frac{\partial \boldsymbol{\mu}_{\mathbf{x}_i}^+}{\partial \boldsymbol{\theta}} \tag{A.1}$$

$$\mathbf{C}_i' = \frac{\partial h_i(\mathbf{x}, \boldsymbol{\mu}_{\boldsymbol{\theta}_i}^+)}{\partial \mathbf{x}} + \frac{\partial h_i(\mathbf{x}, \boldsymbol{\mu}_{\boldsymbol{\theta}_i}^+)}{\partial \boldsymbol{\mu}_{\boldsymbol{\theta}_i}^+} \cdot \frac{\partial \boldsymbol{\mu}_{\boldsymbol{\theta}_i}^+}{\partial \mathbf{x}} \tag{A.2}$$

Inspired by the Kalman filter equations, we find:

$$\boldsymbol{\mu}_{\mathbf{x}_i}^+ = \boldsymbol{\mu}_{\mathbf{x}_i}^- + G_i(\mathbf{y}_i - h_i\left(\boldsymbol{\mu}_{\mathbf{x}_i}^-, \boldsymbol{\theta}\right)) \tag{A.3}$$

$$\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^+ = \boldsymbol{\mu}_{\boldsymbol{\theta}_i}^- + S_i(\mathbf{y}_i - h_i\left(\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^-, \mathbf{x}\right)) \tag{A.4}$$

where $G_i = \mathbf{E}_{\mathbf{x}_i}^- \boldsymbol{C}_i^T \left(\mathbf{C}_i \mathbf{E}_{\mathbf{x}_i}^- \boldsymbol{C}_i^T + \mathbf{I}_{K_i}\right)^{-1}$ and $S_i = \mathbf{E}_{\boldsymbol{\theta}_i}^- \boldsymbol{B}_i^T \left(\mathbf{B}_i \mathbf{E}_{\boldsymbol{\theta}_i}^- \boldsymbol{B}_i^T + \mathbf{I}_{K_i}\right)^{-1}$ are Kalman gains.

Then from (A.3) and (A.4):

$$\frac{\partial \boldsymbol{\mu}_{\mathbf{x}_i}^{+}}{\partial \boldsymbol{\theta}} = \frac{\partial \boldsymbol{\mu}_{\mathbf{x}_i}^{-}}{\partial \boldsymbol{\theta}} - G_i \frac{\partial h_i\left(\boldsymbol{\mu}_{\mathbf{x}_i}^{-}, \boldsymbol{\theta}\right)}{\partial \boldsymbol{\theta}} = \frac{\partial \boldsymbol{\mu}_{\mathbf{x}_i}^{-}}{\partial \boldsymbol{\theta}} - G_i B_i\left(\boldsymbol{\mu}_{\mathbf{x}_i}^{-}\right) \tag{A.5}$$

$$\frac{\partial \boldsymbol{\mu}_{\boldsymbol{\theta}_i}^{+}}{\partial \mathbf{x}} = \frac{\partial \boldsymbol{\mu}_{\boldsymbol{\theta}_i}^{-}}{\partial \mathbf{x}} - S_i \frac{\partial h_i\left(\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^{-}, \mathbf{x}\right)}{\partial \mathbf{x}} = \frac{\partial \boldsymbol{\mu}_{\boldsymbol{\theta}_i}^{-}}{\partial \mathbf{x}} - S_i C_i\left(\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^{-}\right) \tag{A.6}$$

As $\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^{-} = \boldsymbol{\mu}_{\boldsymbol{\theta}_{i-1}}^{+}$, which is not a function of $\mathbf{x}$, so $\frac{\partial \boldsymbol{\mu}_{\boldsymbol{\theta}_i}^{-}}{\partial \mathbf{x}} = 0$. However, $\frac{\partial \boldsymbol{\mu}_{\mathbf{x}_i}^{-}}{\partial \boldsymbol{\theta}}$ is not zero because the state transitional equation for the state vector $\mathbf{x}_i$ is:

$$\boldsymbol{\mu}_{\mathbf{x}_i}^{-} = f_{i-1}\left(\boldsymbol{\mu}_{\mathbf{x}_{i-1}}^{+}, \boldsymbol{\theta}\right) = \mathbf{P}_{i-1}(\boldsymbol{\theta})\, \boldsymbol{\mu}_{\mathbf{x}_{i-1}}^{+} + \mathbf{p}_{i-1}(\boldsymbol{\theta}) \tag{A.7}$$

so:

$$\mathbf{L}_{i-1} = \frac{\partial \boldsymbol{\mu}_{\mathbf{x}_i}^{-}}{\partial \boldsymbol{\theta}} = \frac{\partial f_{i-1}\left(\boldsymbol{\mu}_{\mathbf{x}_{i-1}}^{+}, \boldsymbol{\theta}\right)}{\partial \boldsymbol{\theta}} + \frac{\partial f_{i-1}\left(\boldsymbol{\mu}_{\mathbf{x}_{i-1}}^{+}, \boldsymbol{\theta}\right)}{\partial \boldsymbol{\mu}_{\mathbf{x}_{i-1}}^{+}} \cdot \frac{\partial \boldsymbol{\mu}_{\mathbf{x}_{i-1}}^{+}}{\partial \boldsymbol{\theta}} = \frac{\partial f_{i-1}\left(\boldsymbol{\mu}_{\mathbf{x}_{i-1}}^{+}, \boldsymbol{\theta}\right)}{\partial \boldsymbol{\theta}} + \mathbf{P}_{i-1}(\boldsymbol{\theta}) \cdot \frac{\partial \boldsymbol{\mu}_{\mathbf{x}_{i-1}}^{+}}{\partial \boldsymbol{\theta}} \tag{A.8}$$

where $\frac{\partial \boldsymbol{\mu}_{\mathbf{x}_{i-1}}^{+}}{\partial \boldsymbol{\theta}}$ refers to the result at the (i-1)th time instant.

(2) Predictive PDF for $\mathbf{x}_i$

The predictive PDF for the state vector $\mathbf{x}_i$ can be analytically evaluated:

$$\begin{aligned}
&p\left(\mathbf{x}_i|\mathbf{y}_{0:\,i-1}\right) = \int p\left(\mathbf{x}_i|\boldsymbol{\theta}_{i-1}, \mathbf{y}_{0:\,i-1}\right) p\left(\boldsymbol{\theta}_{i-1}|\mathbf{y}_{0:\,i-1}\right) d\boldsymbol{\theta}_{i-1} \\
&= \iint p\left(\mathbf{x}_i|\mathbf{x}_{i-1,}\boldsymbol{\theta}_{i-1}, \mathbf{y}_{0:\,i-1}\right) p\left(\mathbf{x}_{i-1}|\boldsymbol{\theta}_{i-1}, \mathbf{y}_{0:\,i-1}\right) d\mathbf{x}_{i-1}.p\left(\boldsymbol{\theta}_{i-1}|\mathbf{y}_{0:\,i-1}\right) d\boldsymbol{\theta}_{i-1} \\
&= \int \mathcal{N}\left(\mathbf{x}_i\middle|f_{i-1}\left(\boldsymbol{\mu}_{\mathbf{x}_{i-1}}^{+}, \boldsymbol{\theta}_{i-1}\right), \mathbf{Q}_i + \mathbf{P}_i\left(\boldsymbol{\theta}_{i-1}\right) \mathbf{E}_{\mathbf{x}_{i-1}}^{+} \mathbf{P}_i^{T}\left(\boldsymbol{\theta}_{i-1}\right)\right) .\mathcal{N}\left(\boldsymbol{\theta}_{i-1}\middle|\boldsymbol{\mu}_{\boldsymbol{\theta}_{i-1}}^{+}, \boldsymbol{\Sigma}_{\boldsymbol{\theta}_{i-1}}^{+}\right) d\boldsymbol{\theta}_{i-1} \\
&\approx \int \mathcal{N}\left(\mathbf{x}_i\middle|\mathbf{L}_{i-1}\boldsymbol{\theta}_{i-1} + \left[f_{i-1}\left(\boldsymbol{\mu}_{\mathbf{x}_{i-1}}^{+}, \boldsymbol{\mu}_{\boldsymbol{\theta}_{i-1}}^{+}\right) - \mathbf{L}_{i-1}\boldsymbol{\mu}_{\boldsymbol{\theta}_{i-1}}^{+}\right], \mathbf{Q}_i + \mathbf{P}_i\left(\boldsymbol{\theta}_{i-1}\right) \mathbf{E}_{\mathbf{x}_{i-1}}^{+} \mathbf{P}_i^{T}\left(\boldsymbol{\theta}_{i-1}\right)\right) \\
&\cdot \mathcal{N}\left(\boldsymbol{\theta}_{i-1}\middle|\boldsymbol{\mu}_{\boldsymbol{\theta}_{i-1}}^{+}, \boldsymbol{\Sigma}_{\boldsymbol{\theta}_{i-1}}^{+}\right) d\boldsymbol{\theta}_{i-1} \\
&= \mathcal{N}\left(\mathbf{x}_i\middle|f_{i-1}\left(\boldsymbol{\mu}_{\mathbf{x}_{i-1}}^{+}, \boldsymbol{\mu}_{\boldsymbol{\theta}_{i-1}}^{+}\right), \mathbf{Q}_i + \mathbf{P}_i\left(\boldsymbol{\theta}_{i-1}\right) \mathbf{E}_{\mathbf{x}_{i-1}}^{+} \mathbf{P}_i^{T}\left(\boldsymbol{\theta}_{i-1}\right) + \mathbf{L}_{i-1} \boldsymbol{\Sigma}_{\boldsymbol{\theta}_{i-1}}^{+} \mathbf{L}_{i-1}^{T}\right)
\end{aligned} \tag{A.9}$$

The approximate equation above is based on the Taylor expansion of function $f_{i-1}\left(\boldsymbol{\mu}_{\mathbf{x}_{i-1}}^{+}, \boldsymbol{\theta}_i\right)$ at $\boldsymbol{\mu}_{\boldsymbol{\theta}_{i-1}}^{+}$ by omitting terms of higher order than the first degree:

$$\begin{aligned}
&f_{i-1}\left(\boldsymbol{\mu}_{\mathbf{x}_{i-1}}^{+}, \boldsymbol{\theta}_{i-1}\right) \approx f_{i-1}\left(\boldsymbol{\mu}_{\mathbf{x}_{i-1}}^{+}, \boldsymbol{\mu}_{\boldsymbol{\theta}_{i-1}}^{+}\right) + \frac{\partial f_{i-1}\left(\boldsymbol{\mu}_{\mathbf{x}_{i-1}}^{+}, \boldsymbol{\theta}_{i-1}\right)}{\partial \boldsymbol{\theta}}(\boldsymbol{\theta}_{i-1} - \boldsymbol{\mu}_{\boldsymbol{\theta}_{i-1}}^{+}) \\
&= f_{i-1}\left(\boldsymbol{\mu}_{\mathbf{x}_{i-1}}^{+}, \boldsymbol{\mu}_{\boldsymbol{\theta}_{i-1}}^{+}\right) + \mathbf{L}_{i-1}(\boldsymbol{\theta}_{i-1} - \boldsymbol{\mu}_{\boldsymbol{\theta}_{i-1}}^{+}) \\
&= \mathbf{L}_{i-1}\boldsymbol{\theta}_{i-1} + \left[f_{i-1}\left(\boldsymbol{\mu}_{\mathbf{x}_{i-1}}^{+}, \boldsymbol{\mu}_{\boldsymbol{\theta}_{i-1}}^{+}\right) - \mathbf{L}_{i-1}\boldsymbol{\mu}_{\boldsymbol{\theta}_{i-1}}^{+}\right]
\end{aligned} \tag{A.10}$$

(3) Likelihood functions for $\mathbf{x}_i$ and $\boldsymbol{\theta}_i$

Based on the maximum entropy PDF for the prediction error $\mathbf{e}_i$, one gets a Gaussian likelihood function for $\mathbf{x}_i$ and $\boldsymbol{\theta}_i$ given the measurement $\mathbf{y}_i$ at the ith time instant:

$$p\left(\mathbf{y}_i|\mathbf{x}_i, \boldsymbol{\theta}_i, \beta\right) = \mathcal{N}\left(\mathbf{y}_i|h_i\left(\mathbf{x}_i, \boldsymbol{\theta}_i\right), \beta^{-1}\mathbf{I}_K\right) = \mathcal{N}\left(\mathbf{y}_i|\mathbf{C}_i\mathbf{x}_i + \mathbf{c}_i, \beta^{-1}\mathbf{I}_{K_i}\right) = \mathcal{N}\left(\mathbf{y}_i|\mathbf{B}_i\boldsymbol{\theta}_i + \mathbf{b}_i, \beta^{-1}\mathbf{I}_{K_i}\right) \tag{A.11}$$

For the state vector $\mathbf{x}_i$, the likelihood function is computed as:

$$p\left(\mathbf{y}_i|\mathbf{x}_i, \beta, \mathbf{y}_{0:\,i-1}\right) = \int p\left(\mathbf{y}_i|\boldsymbol{\theta}_i, \mathbf{x}_i, \beta\right) p\left(\boldsymbol{\theta}_i|\mathbf{y}_{0:\,i-1}\right) d\boldsymbol{\theta}_i$$

$$
\begin{aligned}
&= \int \mathcal{N}\left(\mathbf{y}_i|h_i\left(\mathbf{x}_i, \boldsymbol{\theta}_i\right), \beta^{-1}\mathbf{I}_{K_i}\right)\mathcal{N}(\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^-, \boldsymbol{\Sigma}_{\boldsymbol{\theta}_i}^-)d\boldsymbol{\theta}_i \\
&\approx \mathcal{N}\left(\mathbf{y}_i|h_i\left(\mathbf{x}_i, \boldsymbol{\mu}_{\boldsymbol{\theta}_i}^-\right), \beta^{-1}\mathbf{I}_{K_i} + \mathbf{B}_i\left(\mathbf{x}_i\right)\boldsymbol{\Sigma}_{\boldsymbol{\theta}_i}^-\left(\mathbf{B}_i\left(\mathbf{x}_i\right)\right)^T\right) \\
&= \mathcal{N}\left(\mathbf{y}_i|\mathbf{C}_i\left(\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^-\right)\mathbf{x}_i + \mathbf{c}_i\left(\boldsymbol{\mu}_{\boldsymbol{\theta}_i}^-\right), \beta^{-1}\mathbf{I}_{K_i} + \mathbf{B}_i\left(\mathbf{x}_i\right)\boldsymbol{\Sigma}_{\boldsymbol{\theta}_i}^-\left(\mathbf{B}_i\left(\mathbf{x}_i\right)\right)^T\right)
\end{aligned} \tag{A.12}
$$

For the model parameter vector $\boldsymbol{\theta}_i$, the likelihood function is computed as:

$$
\begin{aligned}
&p\left(\mathbf{y}_i|\boldsymbol{\theta}_i, \beta, \mathbf{y}_{0:\,i-1}\right) = \int p\left(\mathbf{y}_i|\boldsymbol{\theta}_i, \mathbf{x}_i, \beta\right)p\left(\mathbf{x}_i|\mathbf{y}_{0:\,i-1}\right)d\mathbf{x}_i \\
&= \int \mathcal{N}\left(\mathbf{y}_i|h_i\left(\mathbf{x}_i, \boldsymbol{\theta}_i\right), \beta^{-1}\mathbf{I}_{K_i}\right)\mathcal{N}(\boldsymbol{\mu}_{\mathbf{x}_i}^-, \boldsymbol{\Sigma}_{\mathbf{x}_i}^-)d\mathbf{x}_i \\
&\approx \mathcal{N}\left(\mathbf{y}_i|h_i\left(\boldsymbol{\mu}_{\mathbf{x}_i}^-, \boldsymbol{\theta}_i\right), \beta^{-1}\mathbf{I}_{K_i} + \mathbf{C}_i\left(\boldsymbol{\theta}_i\right)\boldsymbol{\Sigma}_{\mathbf{x}_i}^-\left(\mathbf{C}_i\left(\boldsymbol{\theta}_i\right)\right)^T\right) \\
&= \mathcal{N}\left(\mathbf{y}_i|\mathbf{B}_i\left(\boldsymbol{\mu}_{\mathbf{x}_i}^-\right)\boldsymbol{\theta}_i + \mathbf{b}_i\left(\boldsymbol{\mu}_{\mathbf{x}_i}^-\right), \beta^{-1}\mathbf{I}_{K_i} + \mathbf{C}_i\left(\boldsymbol{\theta}_i\right)\boldsymbol{\Sigma}_{\mathbf{x}_i}^-\left(\mathbf{C}_i\left(\boldsymbol{\theta}_i\right)\right)^T\right)
\end{aligned} \tag{A.13}
$$

The approximate equations in (A.12) and (A.13) are due to:

$$
h_i\left(\mathbf{x}_i, \boldsymbol{\theta}_i\right) \approx h_i\left(\mathbf{x}_i, \boldsymbol{\mu}_{\boldsymbol{\theta}_i}^-\right) + h_i'\left(\mathbf{x}_i, \boldsymbol{\mu}_{\boldsymbol{\theta}_i}^-\right)\left(\boldsymbol{\theta}_i - \boldsymbol{\mu}_{\boldsymbol{\theta}_i}^-\right) \tag{A.14a}
$$

$$
h_i\left(\mathbf{x}_i, \boldsymbol{\theta}_i\right) \approx h_i\left(\boldsymbol{\mu}_{\mathbf{x}_i}^-, \boldsymbol{\theta}_i\right) + h_i'\left(\boldsymbol{\mu}_{\mathbf{x}_i}^-, \boldsymbol{\theta}_i\right)\left(\mathbf{x}_i - \boldsymbol{\mu}_{\mathbf{x}_i}^-\right) \tag{A.14b}
$$

Appendix B. Online updating of the state vector in the Kalman filter and marginalizing over the "nuisance" noise parameters by Laplace's asymptotic approximation

We assume that the updated PDF for the parameter vector $\mathbf{w}_{i-1} \in \mathbb{R}^{S_{i-1}\times 1}$ at the (i-1)th time instant is given by:

$$
p\left(\mathbf{w}_{i-1}|\mathbf{y}_{0:\,i-1}, \mathbf{w}_{0:\,i-1}^c, \beta\right) = \mathcal{N}\left(\mathbf{w}_{i-1}|\boldsymbol{\mu}_{i-1}, \beta^{-1}\mathbf{E}_{i-1}^+\right) \tag{B.1}
$$

We also assume that the state transition PDF for parameter vector $\mathbf{w}_i \in \mathbb{R}^{S_i\times 1}$ at the ith time instant has the following generic Gaussian form:

$$
p\left(\mathbf{w}_i|\mathbf{w}_{i-1}, \boldsymbol{\gamma}_i, \beta\right) = \mathcal{N}\left(\mathbf{w}_i|\mathbf{H}\mathbf{w}_{i-1} + \mathbf{h}, \beta^{-1}\boldsymbol{\Omega}_i\right) \tag{B.2}
$$

where $\boldsymbol{\gamma}_i = [\gamma_{i,1}, \ldots, \gamma_{i,S_i}]^T$, $\boldsymbol{\Omega}_i = \mathrm{diag}\left(r_{i,1}, \ldots, r_{i,S_i}\right)$.

Then the predictive PDF for the parameter vector $\mathbf{w}_i$ is:

$$
\begin{aligned}
&p\left(\mathbf{w}_i|\mathbf{y}_{0:\,i-1}, \mathbf{w}_{0:\,i-1}^c, \boldsymbol{\gamma}_i, \beta\right) = \int p\left(\mathbf{w}_i|\mathbf{w}_{i-1}, \boldsymbol{\gamma}_i, \beta\right)p\left(\mathbf{w}_{i-1}|\mathbf{y}_{0:\,i-1}, \mathbf{w}_{0:\,i-1}^c, \beta\right)d\mathbf{w}_{i-1} \\
&= \mathcal{N}\left(\mathbf{w}_i|\boldsymbol{\mu}_i^-, \boldsymbol{\Sigma}_i^-\right) = \mathcal{N}\left(\mathbf{w}_i|\boldsymbol{\mu}_i^-, \beta^{-1}\mathbf{E}_i^-\right) = \mathcal{N}\left(\mathbf{w}_i|\mathbf{H}\boldsymbol{\mu}_{i-1} + \mathbf{h}, \beta^{-1}\left(\boldsymbol{\Omega}_i + \mathbf{H}\mathbf{E}_{i-1}^+\mathbf{H}^T\right)\right)
\end{aligned} \tag{B.3}
$$

Based on two measurement vectors $\mathbf{y}_i \in \mathbb{R}^{K_i\times 1}$ and $\mathbf{w}_i^c \in \mathbb{R}^{S_i\times 1}$, it is assumed that the likelihood functions for $\mathbf{w}_i$ have the following generic Gaussian forms:

$$
p\left(\mathbf{y}_i|\mathbf{w}_i, \beta\right) = \mathcal{N}\left(\mathbf{y}_i|\mathbf{K}\mathbf{w}_i + \mathbf{k}, \beta^{-1}\left(\mathbf{I}_{K_i} + \mathbf{X}\right)\right) \tag{B.4}
$$

$$
p\left(\mathbf{w}_i^c|\mathbf{w}_i, \boldsymbol{\alpha}_i\right) = \mathcal{N}\left(\mathbf{w}_i^c|\mathbf{w}_i, \mathbf{A}_i\right) \tag{B.5}
$$

where: $\boldsymbol{\alpha}_i = [\alpha_{i,1}, \ldots, \alpha_{i,S_i}]^T$, $\mathbf{A}_i = \mathrm{diag}\left(\boldsymbol{\alpha}_{i,1}, \ldots, \boldsymbol{\alpha}_{i,S_i}\right)$.

Based on the specified values of hyper-parameters $\boldsymbol{\gamma}_i$ and β, the posterior PDF of $\mathbf{w}_i$ given the data $\mathbf{y}_i$ is computed by Bayes' theorem:

$$
\begin{aligned}
&p\left(\mathbf{w}_i|\mathbf{y}_{0:\,i}, \mathbf{w}_{0:\,i-1}^c, \boldsymbol{\gamma}_i, \beta\right) = p\left(\mathbf{y}_i|\mathbf{w}_i, \beta\right)p\left(\mathbf{w}_i|\mathbf{y}_{0:\,i-1}, \mathbf{w}_{0:\,i-1}^c, \boldsymbol{\gamma}_i, \beta\right)/p\left(\mathbf{y}_i|\mathbf{y}_{0:\,i-1}, \mathbf{w}_{0:\,i-1}^c, \boldsymbol{\gamma}_i, \beta\right) \\
&= \mathcal{N}\left(\mathbf{w}_i|\boldsymbol{\mu}_i^{+(1)}, \boldsymbol{\Sigma}_i^{+(1)}\right)
\end{aligned} \tag{B.6}
$$

where:

$$\boldsymbol{\mu}_i^{+(1)} = \boldsymbol{\Sigma}_i^{+(1)} \left(\beta \mathbf{K}^T \left(\mathbf{I}_K + \mathbf{X} \right)^{-1} \left(\mathbf{y}_i - \mathbf{k} \right) + \left(\boldsymbol{\Sigma}_i^- \right)^{-1} \boldsymbol{\mu}_i^- \right) = \mathbf{E}_i^{+(1)} \left(\mathbf{K}^T \left(\mathbf{I}_K + \mathbf{X} \right)^{-1} \left(\mathbf{y}_i - \mathbf{k} \right) + \left(\mathbf{E}_i^- \right)^{-1} \boldsymbol{\mu}_i^- \right); \tag{B.7}$$

$$\boldsymbol{\Sigma}_i^{+(1)} = \beta^{-1} \mathbf{E}_i^{+(1)} = \beta^{-1} \left(\left(\mathbf{E}_i^- \right)^{-1} + \mathbf{K}^T \left(\mathbf{I}_{K_i} + \mathbf{X} \right) \mathbf{K} \right)^{-1}; \tag{B.8}$$

$$\begin{aligned} p\left(\mathbf{y}_i | \mathbf{y}_{0:\,i-1}, \mathbf{w}_{0:\,i-1}^c, \boldsymbol{\gamma}_i, \beta \right) &= \int p\left(\mathbf{y}_i | \mathbf{w}_i, \beta \right) p\left(\mathbf{w}_i | \mathbf{y}_{0:\,i-1}, \mathbf{w}_{0:\,i-1}^c, \boldsymbol{\gamma}_i, \beta \right) d\mathbf{w}_i \\ &= \mathcal{N}\left(\mathbf{y}_i | \mathbf{K}\boldsymbol{\mu}_i^- + \mathbf{k}, \beta^{-1} \left(\mathbf{I}_{K_i} + \mathbf{X} + \mathbf{K}\mathbf{E}_i^- \mathbf{K}^T \right) \right) \end{aligned} \tag{B.9}$$

We assume that the posterior $p\left(\boldsymbol{\gamma}_i, \beta | \mathbf{y}_{0:\,i}, \mathbf{w}_{0:\,i-1}^c \right)$ in (B.6) has a unique global maximum over $\boldsymbol{\gamma}_i$ and β, and then we treat $\boldsymbol{\gamma}_i$ and β as 'nuisance' parameter vectors and integrate them out using Laplace's asymptotic approximation:

$$\begin{aligned} p\left(\mathbf{w}_i | \mathbf{y}_{0:\,i}, \mathbf{w}_{0:\,i-1}^c \right) &= \int p\left(\mathbf{w}_i | \mathbf{y}_{0:\,i}, \mathbf{w}_{0:\,i-1}^c, \boldsymbol{\gamma}_i, \beta \right) p\left(\boldsymbol{\gamma}_i, \beta | \mathbf{y}_{0:\,i}, \mathbf{w}_{0:\,i-1}^c \right) d\boldsymbol{\gamma}_i d\beta \\ &\approx p\left(\mathbf{w}_i | \mathbf{y}_{0:\,i}, \mathbf{w}_{0:\,i-1}^c, \tilde{\boldsymbol{\gamma}}_i, \tilde{\beta} \right) \end{aligned} \tag{B.10}$$

where

$$\begin{aligned} \left\{ \tilde{\boldsymbol{\gamma}}_i, \tilde{\beta} \right\} &= \arg\max\, p\left(\boldsymbol{\gamma}_i, \beta | \mathbf{y}_{0:\,i}, \mathbf{w}_{0:\,i-1}^c \right) \\ &= \operatorname{argmax} \left\{ p\left(\mathbf{y}_i | \mathbf{y}_{0:\,i-1}, \mathbf{w}_{0:\,i-1}^c, \boldsymbol{\gamma}_i, \beta \right)\ p\left(\boldsymbol{\gamma}_i, \beta | \mathbf{y}_{0:\,i-1}, \mathbf{w}_{0:\,i-1}^c \right) \right\} \\ &= \operatorname{argmax} \left\{ p\left(\mathbf{y}_i | \mathbf{y}_{0:\,i-1}, \mathbf{w}_{0:\,i-1}^c, \boldsymbol{\gamma}_i, \beta \right)\ p\left(\boldsymbol{\gamma}_i, \boldsymbol{\alpha}_i \right) \right\} \end{aligned} \tag{B.11}$$

To find these MAP values, we first differentiate the logarithm function of $p\left(\boldsymbol{\gamma}_i, \beta | \mathbf{y}_{0:\,i}, \mathbf{w}_{0:\,i-1}^c \right)$ with respect to $\gamma_{i,j}$. To proceed, we first define $\boldsymbol{\Gamma}_i = \mathbf{K}\mathbf{H}\mathbf{E}_{i-1}\mathbf{H}^T\mathbf{K}^T + \mathbf{I}_{K_i} + \mathbf{X}$, $\mathbf{Z}_i = \left(\mathbf{K}^T \boldsymbol{\Gamma}_i^{-1} \mathbf{K} + \boldsymbol{\Omega}_i^{-1} \right)^{-1}$ and $\boldsymbol{\kappa}_i = \mathbf{Z}_i \left(\boldsymbol{\Omega}_i^{-1} \boldsymbol{\mu}_i^- + \mathbf{K}^T \boldsymbol{\Gamma}_i^{-1} \left(\mathbf{y}_i - \mathbf{k} \right) \right)$. Ignoring terms in the logarithm function of $p\left(\boldsymbol{\gamma}_i, \beta | \mathbf{y}_{1:\,i}, \mathbf{w}_{1:\,i-1}^c \right)$ that are independent of $\boldsymbol{\gamma}_i$, the objective function is:

$$\begin{aligned} \mathcal{L}\left(\boldsymbol{\gamma}_i \right) &= -\frac{1}{2} \log \left| \left(\boldsymbol{\Gamma}_i + \mathbf{K}\boldsymbol{\Omega}_i\mathbf{K}^T \right) \right| - \frac{1}{2} \beta \left(\mathbf{y}_i - \mathbf{K}\boldsymbol{\mu}_i^- - \mathbf{k} \right)^T \left(\boldsymbol{\Gamma}_i + \mathbf{K}\boldsymbol{\Omega}_i\mathbf{K}^T \right)^{-1} \left(\mathbf{y}_i - \mathbf{K}\boldsymbol{\mu}_i^- - \mathbf{k} \right) \\ &= -\frac{1}{2} \left(\log |\boldsymbol{\Gamma}_i| + \log |\boldsymbol{\Omega}_i| - \log |\mathbf{Z}_i| \right) - \frac{1}{2} \beta \Big[\left(\mathbf{y}_i - \mathbf{K}\boldsymbol{\kappa}_i - \mathbf{k} \right)^T \boldsymbol{\Gamma}_i^{-1} \left(\mathbf{y}_i - \mathbf{K}\boldsymbol{\kappa}_i - \mathbf{k} \right) \\ &\quad + \left(\boldsymbol{\kappa}_i - \boldsymbol{\mu}_i^- \right)^T \boldsymbol{\Omega}_i^{-1} \left(\boldsymbol{\kappa}_i - \boldsymbol{\mu}_i^- \right) \Big] \end{aligned} \tag{B.12}$$

Based on the determinant identity and Woodbury matrix identity, respectively,

$$\log \left| \boldsymbol{\Gamma}_i + \mathbf{K}\boldsymbol{\Omega}_i\mathbf{K}^T \right| = \log |\boldsymbol{\Gamma}_i| + \log |\boldsymbol{\Omega}_i| + \log \left| \mathbf{K}^T \boldsymbol{\Gamma}_i^{-1} \mathbf{K} + \boldsymbol{\Omega}_i^{-1} \right| = \log |\boldsymbol{\Gamma}_i| + \log |\boldsymbol{\Omega}_i| - \log |\mathbf{Z}_i| \tag{B.13}$$

$$\begin{aligned} &\left(\mathbf{y}_i - \mathbf{K}\boldsymbol{\mu}_i^- - \mathbf{k} \right)^T \left(\boldsymbol{\Gamma}_i + \mathbf{K}\boldsymbol{\Omega}_i\mathbf{K}^T \right)^{-1} \left(\mathbf{y}_i - \mathbf{K}\boldsymbol{\mu}_i^- - \mathbf{k} \right) \\ &= \left(\mathbf{y}_i - \mathbf{K}\boldsymbol{\kappa}_i - \mathbf{k} \right)^T \boldsymbol{\Gamma}_i^{-1} \left(\mathbf{y}_i - \mathbf{K}\boldsymbol{\kappa}_i - \mathbf{b} \right) + \left(\boldsymbol{\kappa}_i - \boldsymbol{\mu}_i^- \right)^T \boldsymbol{\Omega}_i^{-1} \left(\boldsymbol{\kappa}_i - \boldsymbol{\mu}_i^- \right) \end{aligned} \tag{B.14}$$

Thus, the derivative of $\mathcal{L}\left(\boldsymbol{\gamma}_i \right)$ in (B.12) with respect to $\gamma_{i,j}$ is given by:

$$\frac{\partial \mathcal{L}\left(\boldsymbol{\gamma}_i \right)}{\partial \gamma_{i,j}} = -\frac{1}{2} \left[\frac{\partial \log |\boldsymbol{\Omega}_i|}{\partial \gamma_{i,j}} - \frac{\partial \log \left| \mathbf{Z}_i^{-1} \right|}{\partial \gamma_{i,j}} + \frac{\partial \left[\beta \left(\boldsymbol{\kappa}_i - \boldsymbol{\mu}_i^- \right)^T \boldsymbol{\Omega}_i^{-1} \left(\boldsymbol{\kappa}_i - \boldsymbol{\mu}_i^- \right) \right]}{\partial \gamma_{i,j}} \right] \tag{B.15}$$

If the derivative of matrix $\boldsymbol{\Omega}$ with respect to component α_j is $\frac{\partial \boldsymbol{\Omega}_i(\boldsymbol{\gamma}_i)}{\partial \gamma_{i,j}} = \boldsymbol{\Pi}_j$, then we have $\boldsymbol{\Omega}_i^{-1}\boldsymbol{\Pi}_j = \gamma_{i,j}^{-1}\boldsymbol{\Pi}_j$ and $\boldsymbol{\Omega}_i^{-2}\boldsymbol{\Pi}_j = \gamma_{i,j}^{-2}\boldsymbol{\Pi}_j$. Setting the derivatives to zero leads to the update formulas for the parameters $\gamma_{i,j}$:

$$\frac{\partial \mathcal{L}\left(\boldsymbol{\gamma}_i \right)}{\partial \gamma_{i,j}} = -\frac{1}{2} \left[\frac{\partial \log |\boldsymbol{\Omega}_i|}{\partial \gamma_{i,j}} - \frac{\partial \log \left| \mathbf{Z}_i^{-1} \right|}{\partial \gamma_{i,j}} + \frac{\partial \left[\beta \left(\boldsymbol{\kappa}_i - \boldsymbol{\mu}_i^- \right)^T \boldsymbol{\Omega}_i^{-1} \left(\boldsymbol{\kappa}_i - \boldsymbol{\mu}_i^- \right) \right]}{\partial \gamma_{i,j}} \right]$$

$$= \mathrm{tr}\left(\boldsymbol{\Omega}_i^{-1}\boldsymbol{\Pi}_j\right) - \mathrm{tr}\left(\mathbf{Z}_i\boldsymbol{\Omega}_i^{-2}\boldsymbol{\Pi}_j\right) - \beta\left(\boldsymbol{\kappa}_i - \boldsymbol{\mu}_i^-\right)^T\boldsymbol{\Omega}_i^{-2}\boldsymbol{\Pi}_j\left(\boldsymbol{\kappa}_i - \boldsymbol{\mu}_i^-\right)$$
$$= \gamma_{i,j}^{-1}\mathrm{tr}\left(\boldsymbol{\Pi}_j\right) - \gamma_{i,j}^{-2}\mathrm{tr}\left(\mathbf{Z}_i\boldsymbol{\Pi}_j\right) - \beta\gamma_{i,j}^{-2}\left(\boldsymbol{\kappa}_i - \boldsymbol{\mu}_i^-\right)^T\boldsymbol{\Pi}_j\left(\boldsymbol{\kappa}_i - \boldsymbol{\mu}_i^-\right) \tag{B.16}$$

The MAP estimate of $\gamma_{i,j}$ is derived as:

$$\tilde{\gamma}_{i,j} = \mathrm{tr}\left(\mathbf{Z}_i\boldsymbol{\Pi}_j\right) + \beta\left(\boldsymbol{\kappa}_i - \mathbf{K}\boldsymbol{\mu}_i^-\right)^T\boldsymbol{\Pi}_j\left(\boldsymbol{\kappa}_i - \mathbf{K}\boldsymbol{\mu}_i^-\right) \tag{B.17}$$

Note that when differentiating (B.14) with respect to uncertain parameters $\gamma_{i,j}$, the terms involving derivatives of $\boldsymbol{\kappa}_i$ drop out since:

$$\begin{aligned}
&-(d\boldsymbol{\kappa}_i)^T\mathbf{K}^T\boldsymbol{\Gamma}_i^{-1}\left(\mathbf{y}_i - \mathbf{K}\boldsymbol{\kappa}_i - \mathbf{k}\right) - \left(\mathbf{y}_i - \mathbf{K}\boldsymbol{\kappa}_i - \mathbf{k}\right)^T\boldsymbol{\Gamma}_i^{-1}\mathbf{K}\cdot d\boldsymbol{\kappa}_i + (d\boldsymbol{\kappa}_i)^T\boldsymbol{\Omega}_i^{-1}\left(\boldsymbol{\kappa}_i - \boldsymbol{\mu}_i^-\right)\\
&+\left(\boldsymbol{\kappa}_i - \boldsymbol{\mu}_i^-\right)^T\boldsymbol{\Omega}_i^{-1}\cdot(d\boldsymbol{\kappa}_i)\\
&= (d\boldsymbol{\kappa}_i)^T\left(\mathbf{K}^T\boldsymbol{\Gamma}_i^{-1}\mathbf{K} + \boldsymbol{\Omega}_i^{-1}\right)\boldsymbol{\kappa}_i + \boldsymbol{\kappa}_i^T\left(\mathbf{K}^T\boldsymbol{\Gamma}_i^{-1}\mathbf{K} + \boldsymbol{\Omega}_i^{-1}\right)d\boldsymbol{\kappa}_i\\
&-(d\boldsymbol{\kappa}_i)^T\mathbf{K}^T\boldsymbol{\Gamma}_i^{-1}\left(\mathbf{y}_i - \mathbf{k}\right) - \left(\mathbf{y}_i - \mathbf{k}\right)^T\boldsymbol{\Gamma}_i^{-1}\mathbf{K}\cdot d\boldsymbol{\kappa}_i - (d\boldsymbol{\kappa}_i)^T\boldsymbol{\Omega}_i^{-1}\boldsymbol{\mu}_i^- - \left(\boldsymbol{\mu}_i^-\right)^T\boldsymbol{\Omega}_i^{-1}\cdot d\boldsymbol{\kappa}_i\\
&= (d\boldsymbol{\kappa}_i)^T\mathbf{Z}_i^{-1}\boldsymbol{\kappa}_i + \boldsymbol{\kappa}_i^T\mathbf{Z}_i^{-1}\cdot d\boldsymbol{\kappa}_i - (d\boldsymbol{\kappa}_i)^T\mathbf{Z}_i^{-1}\boldsymbol{\kappa}_i - \boldsymbol{\kappa}_i^T\mathbf{Z}_i^{-1}\cdot d\boldsymbol{\kappa}_i = 0
\end{aligned} \tag{B.18}$$

We then infer the posterior PDF for β, which is readily obtained by Bayes' theorem:

$$\begin{aligned}
&p\left(\beta|\mathbf{y}_{0:i}, \mathbf{w}_{0:i-1}^c, \boldsymbol{\gamma}_i, \boldsymbol{\alpha}_i\right) \propto p\left(\mathbf{y}_i|\mathbf{y}_{0:i-1}, \mathbf{w}_{0:i-1}^c, \boldsymbol{\gamma}_i, \beta\right)p\left(\beta|\mathbf{y}_{0:i-1}, \mathbf{w}_{0:i-1}^c\right)\\
&= \frac{\beta^{\left(a_{i-1}-1+K_i/2\right)}}{(2\pi)^{K_i/2}}\frac{b_{i-1}^{a_{i-1}}}{\Gamma\left(a_{i-1}\right)}|\mathbf{B}|^{-\frac{1}{2}}\exp\left\{-\frac{\beta}{2}\left(\left(\mathbf{y}_i - \mathbf{K}\boldsymbol{\mu}_i^- - \mathbf{k}\right)^T\left(\mathbf{I}_{K_i} + \mathbf{X} + \mathbf{K}\mathbf{E}_i^-\mathbf{K}^T\right)^{-1}\left(\mathbf{y}_i - \mathbf{K}\boldsymbol{\mu}_i^- - \mathbf{k}\right)\right)\right\}\\
&\propto \mathrm{Gamma}\left(\beta|a_i, b_i\right)
\end{aligned} \tag{B.19}$$

where the Gamma distribution has shape and rate parameters:

$$a_i = a_{i-1} + K_i/2 \tag{B.20}$$

$$b_i = b_{i-1} + \frac{1}{2}\left[\left(\mathbf{y}_i - \mathbf{K}\boldsymbol{\mu}_i^- - \mathbf{b}\right)^T\left(\mathbf{I}_{K_i} + \mathbf{X} + \mathbf{K}\mathbf{E}_i^-\mathbf{K}^T\right)^{-1}\left(\mathbf{y}_i - \mathbf{K}\boldsymbol{\mu}_i^- - \mathbf{b}\right)\right] \tag{B.21}$$

The MAP value for β given the data up to the ith time instant is given by:

$$\tilde{\beta} = a_i/b_i \tag{B.22}$$

We then treat the posterior having observed $\mathbf{y}_i$ as the 'prior' for the remaining data $\mathbf{w}_i^c$ and obtain the final posterior result based on Bayes' theorem:

$$\begin{aligned}
&p\left(\mathbf{w}_i|\mathbf{y}_{0:i}, \mathbf{w}_{0:i}^c, \boldsymbol{\alpha}_i\right) = p\left(\mathbf{w}_i^c|\mathbf{w}_i, \boldsymbol{\alpha}_i\right)p\left(\mathbf{w}_i|\mathbf{y}_{0:i}, \mathbf{w}_{0:i-1}^c\right)/p\left(\mathbf{w}_i^c|\mathbf{y}_{0:i}, \mathbf{w}_{0:i-1}^c, \boldsymbol{\alpha}_i\right)\\
&= \mathcal{N}\left(\mathbf{w}_i|\boldsymbol{\mu}_i^{+(2)}, \boldsymbol{\Sigma}_i^{+(2)}\right)
\end{aligned} \tag{B.23}$$

where:

$$\boldsymbol{\Sigma}_i^{+(2)} = \left(\mathbf{A}_i^{-1} + \left(\boldsymbol{\Sigma}_i^{+(1)}\right)^{-1}\right)^{-1} \tag{B.24a}$$

$$\boldsymbol{\mu}_i^{+(2)} = \boldsymbol{\Sigma}_i^{+(2)}\left(\mathbf{A}_i^{-1}\mathbf{w}_i^c + \left(\boldsymbol{\Sigma}_i^{+(1)}\right)^{-1}\boldsymbol{\mu}_i^{(1)}\right) \tag{B.24b}$$

$$p\left(\mathbf{w}_i^c|\mathbf{y}_{0:i}, \boldsymbol{\alpha}_i\right) = \int p\left(\mathbf{w}_i^c|\mathbf{w}_i, \boldsymbol{\alpha}_i\right)p\left(\mathbf{w}_i|\mathbf{y}_{0:i}, \mathbf{w}_{0:i-1}^c\right)d\mathbf{w}_i = \mathcal{N}\left(\mathbf{w}_i^c|\boldsymbol{\mu}_i^{+(1)}, \mathbf{A}_i + \boldsymbol{\Sigma}_i^{+(1)}\right) \tag{B.24c}$$

It is assumed that the posterior $p\left(\boldsymbol{\alpha}_i|\mathbf{y}_{0:i}, \mathbf{w}_{0:i}^c\right)$ in (B.23) has a unique global maximum over $\boldsymbol{\alpha}_i$ and we integrate it out using Laplace's asymptotic approximation:

$$p\left(\mathbf{w}_i|\mathbf{y}_{0:i}, \mathbf{w}_i^c\right) = \int p\left(\mathbf{w}_i|\mathbf{y}_{0:i}, \mathbf{w}_{0:i}^c, \boldsymbol{\alpha}_i\right)p\left(\boldsymbol{\alpha}_i|\mathbf{y}_{0:i}, \mathbf{w}_i^c\right)d\boldsymbol{\alpha}_i \approx p\left(\mathbf{w}_i|\mathbf{y}_{0:i}, \mathbf{w}_i^c, \widetilde{\boldsymbol{\alpha}}_i\right) \tag{B.25}$$

where

$$\widetilde{\boldsymbol{\alpha}}_i = \arg\max p\left(\boldsymbol{\alpha}_i|\mathbf{y}_{0:i}, \mathbf{w}_{0:i}^c\right)$$

$$= \text{argmax}\left\{p\left(\mathbf{w}_i^c|\mathbf{y}_{0:\,i}, \mathbf{w}_{0:\,i-1}^c, \boldsymbol{\alpha}_i\right)\ p\left(\boldsymbol{\alpha}_i|\mathbf{y}_{0:\,i}, \mathbf{w}_{0:\,i-1}^c\right)\right\}$$
$$= \text{argmax} p\left(\mathbf{w}_i^c|\mathbf{y}_{0:\,i}, \mathbf{w}_{0:\,i-1}^c, \boldsymbol{\alpha}_i\right) \tag{B.26}$$

since broad uniform priors on $\boldsymbol{\alpha}_i$ are assigned. Ignoring terms in the logarithm function of the PDF $p\left(\boldsymbol{\alpha}_i|\mathbf{y}_{0:\,i}, \mathbf{w}_{0:\,i}^c\right)$ that are independent of $\boldsymbol{\alpha}_i$, the objective function is defined as:

$$\mathcal{G}\left(\boldsymbol{\alpha}_i\right) = -\frac{1}{2}\log\left|\mathbf{A}_i + \boldsymbol{\Sigma}_i^{+(1)}\right| - \frac{1}{2}\left(\mathbf{w}_i^c - \boldsymbol{\mu}_i^{+(1)}\right)^T\left(\mathbf{A}_i + \boldsymbol{\Sigma}_i^{+(1)}\right)^{-1}\left(\mathbf{w}_i^c - \boldsymbol{\mu}_i^{+(1)}\right) \tag{B.27}$$

Using the determinant identity and Woodbury matrix identity, respectively, we obtain the following equations:

$$\log\left|\mathbf{A}_i + \boldsymbol{\Sigma}_i^{+(1)}\right| = \log\left(\left|\mathbf{A}_i\right|\left|\boldsymbol{\Sigma}_i^{+(1)}\right|\left|\mathbf{A}_i^{-1} + \left(\boldsymbol{\Sigma}_i^{+(1)}\right)^{-1}\right|\right) = \log\left|\mathbf{A}_i\right| + \log\left|\boldsymbol{\Sigma}_i^{+(1)}\right| + \log\left|\left(\boldsymbol{\Sigma}_i^{+(2)}\right)^{-1}\right| \tag{B.28}$$

$$\left(\mathbf{A}_i + \boldsymbol{\Sigma}_i^{+(1)}\right)^{-1} = \mathbf{A}_i^{-1} - \mathbf{A}_i^{-1}\boldsymbol{\Sigma}_i^{+(2)}\left(\mathbf{A}_i^{-1}\right)^T \tag{B.29}$$

Using the mean and covariance matrix in (B.24b) and (B.24a), we get:

$$\begin{aligned}
&\left(\mathbf{w}_i^c - \boldsymbol{\mu}_i^{+(1)}\right)^T\left(\mathbf{A}_i + \boldsymbol{\Sigma}_i^{+(1)}\right)^{-1}\left(\mathbf{w}_i^c - \boldsymbol{\mu}_i^{+(1)}\right)\\
&= \left(\mathbf{w}_i^c - \boldsymbol{\mu}_i^{+(1)}\right)^T\left(\mathbf{A}_i^{-1} - \mathbf{A}_i^{-1}\boldsymbol{\Sigma}_i^{+(2)}\left(\mathbf{A}_i^{-1}\right)^T\right)\left(\mathbf{w}_i^c - \boldsymbol{\mu}_i^{+(1)}\right)\\
&= \left(\mathbf{w}_i^c - \boldsymbol{\mu}_i^{+(2)}\right)^T\mathbf{A}_i^{-1}\left(\mathbf{w}_i^c - \boldsymbol{\mu}_i^{+(2)}\right) + \left(\boldsymbol{\mu}_i^{+(2)} - \boldsymbol{\mu}_i^{+(1)}\right)^T\left(\boldsymbol{\Sigma}_i^{+(1)}\right)^{-1}\left(\boldsymbol{\mu}_i^{+(2)} - \boldsymbol{\mu}_i^{+(1)}\right)
\end{aligned} \tag{B.30}$$

Using (B.28) and (B.29), the derivative of $\mathcal{G}\left(\boldsymbol{\alpha}_i\right)$ in (B.27) with respect to $\alpha_{i,j}$ is given by:

$$\begin{aligned}
\frac{\partial\mathcal{G}\left(\boldsymbol{\alpha}_i\right)}{\partial\alpha_{i,j}} &= -\frac{1}{2}\frac{\partial}{\partial\alpha_{i,j}}\left[\log\left|\mathbf{A}_i + \boldsymbol{\Sigma}_i^{+(1)}\right| + \beta^{-1}\left(\mathbf{w}_i^c - \boldsymbol{\mu}_i^{+(1)}\right)^T\left(\mathbf{A}_i + \boldsymbol{\Sigma}_i^{+(1)}\right)^{-1}\left(\mathbf{w}_i^c - \boldsymbol{\mu}_i^{+(1)}\right)\right]\\
&= -\frac{1}{2}\left[\frac{\partial\log\left|\mathbf{A}_i\right|}{\partial\alpha_{i,j}} + \frac{\partial\log\left|\left(\boldsymbol{\Sigma}_i^{+(2)}\right)^{-1}\right|}{\partial\alpha_{i,j}} + \left(\mathbf{w}_i^c - \boldsymbol{\mu}_i^{+(2)}\right)^T\frac{\partial\mathbf{A}_i^{-1}}{\partial\alpha_{i,j}}\left(\mathbf{w}_i^c - \boldsymbol{\mu}_i^{+(2)}\right)\right]\\
&= \alpha_{i,j}^{-1} - \alpha_{i,j}^{-2}\left(\boldsymbol{\Sigma}_i^{+(2)}\right)_{jj} - \alpha_{i,j}^{-2}\left(\mathbf{w}_i^c - \boldsymbol{\mu}_i^{+(2)}\right)_j^2
\end{aligned} \tag{B.31}$$

$$\text{Then: } \tilde{\alpha}_{i,j} = \left(\boldsymbol{\Sigma}_i^{+(2)}\right)_{jj} + \left(\mathbf{w}_i^c - \boldsymbol{\mu}_i^{+(2)}\right)_j^2 \tag{B.32}$$

Note that when differentiating (B.30) with respect to uncertain parameters $\alpha_{i,j}$, the terms involving derivatives of $\boldsymbol{\mu}_i^{+(2)}$ drop out since:

$$\begin{aligned}
&-\left(d\boldsymbol{\mu}_i^{+(2)}\right)^T\mathbf{A}_i^{-1}\left(\mathbf{w}_i^c - \boldsymbol{\mu}_i^{+(2)}\right) - \left(\mathbf{w}_i^c - \boldsymbol{\mu}_i^{+(2)}\right)^T\mathbf{A}_i^{-1}\cdot d\boldsymbol{\mu}_i^{+(2)} + \left(d\boldsymbol{\mu}_i^{+(2)}\right)^T\left(\boldsymbol{\Sigma}_i^{+(1)}\right)^{-1}\left(\boldsymbol{\mu}_i^{+(2)} - \boldsymbol{\mu}_i^{+(1)}\right)\\
&+\left(\boldsymbol{\mu}_i^{+(2)} - \boldsymbol{\mu}_i^{+(1)}\right)^T\left(\boldsymbol{\Sigma}_i^{+(1)}\right)^{-1}\cdot\left(d\boldsymbol{\mu}_i^{+(2)}\right)\\
&= \left(d\boldsymbol{\mu}_i^{+(2)}\right)^T\left(\mathbf{A}_i^{-1} + \left(\boldsymbol{\Sigma}_i^{+(1)}\right)^{-1}\right)\boldsymbol{\mu}_i^{+(2)} + \left(\boldsymbol{\mu}_i^{+(2)}\right)^T\left(\mathbf{A}_i^{-1} + \left(\boldsymbol{\Sigma}_i^{+(1)}\right)^{-1}\right)d\boldsymbol{\mu}_i^{+(2)}\\
&-\left(d\boldsymbol{\mu}_i^{+(2)}\right)^T\mathbf{A}_i^{-1}\mathbf{w}_i^c - \left(\mathbf{w}_i^c\right)^T\mathbf{A}_i^{-1}\cdot d\boldsymbol{\mu}_i^{+(2)} - \left(d\boldsymbol{\mu}_i^{+(2)}\right)^T\left(\boldsymbol{\Sigma}_i^{+(1)}\right)^{-1}\boldsymbol{\mu}_i^{+(1)} - \left(\boldsymbol{\mu}_i^{+(1)}\right)^T\left(\boldsymbol{\Sigma}_i^{+(1)}\right)^{-1}\cdot d\boldsymbol{\mu}_i^{+(2)}\\
&= \left(d\boldsymbol{\mu}_i^{+(2)}\right)^T\left(\boldsymbol{\Sigma}_i^{+(2)}\right)^{-1}\boldsymbol{\mu}_i^{+(2)} + \left(\boldsymbol{\mu}_i^{+(2)}\right)^T\left(\boldsymbol{\Sigma}_i^{+(2)}\right)^{-1}\cdot d\boldsymbol{\mu}_i^{+(2)}\\
&-\left(d\boldsymbol{\mu}_i^{+(2)}\right)^T\left(\boldsymbol{\Sigma}_i^{+(2)}\right)^{-1}\boldsymbol{\mu}_i^{+(2)} - \left(\boldsymbol{\mu}_i^{+(2)}\right)^T\left(\boldsymbol{\Sigma}_i^{+(2)}\right)^{-1}\cdot d\boldsymbol{\mu}_i^{+(2)} = 0
\end{aligned} \tag{B.33}$$

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